



KiyaAI Credit Assessment System

BRD_V.1.0

**Submitted to – Commercial Bank of
Kuwait**

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SIGN-OFF

Business Requirements Document (BRD) – KiyaAI Credit Assessment System

This document has been reviewed and approved by the stakeholders listed below.
Their signatures confirm that:

- All requirements stated in the BRD are understood and accepted
- Scope, modules, workflows, and features are agreed upon
- Any changes after sign-off will follow the formal change-management process

Stakeholders – KiyaAI (Solution Provider) & CBK (Client Team)

Organization	Name	Designation	Department / Role	Signature	Date
KiyaAI					
KiyaAI					
KiyaAI					
KiyaAI					
CBK					
CBK					
CBK					
CBK					
CBK					

✓ Approval Statement

By signing this document, all parties acknowledge that the requirements described herein are complete to the best of their knowledge and accurately represent the expected functionality of the **KiyaAI Credit Assessment System**.

Any modifications post-sign-off must go through the agreed change-control process.

1. INTRODUCTION

The **KiyaAI Credit Assessment System** is a comprehensive digital platform designed to automate and enhance the credit evaluation process for financial institutions. The system provides a structured workflow for assessing borrowers by leveraging configurable rating templates, parameter-based scoring, financial analysis, automated ratio calculations, OCR-based data ingestion, and AI-driven insights.

The platform supports end-to-end credit assessment activities, including data collection, template management, parameter configuration, risk scoring, report generation, user management, system configuration, and audit tracking. It ensures consistency, transparency, and regulatory compliance in the credit evaluation lifecycle.

This Business Requirements Document (BRD) defines the detailed functional, non-functional, operational, and user-interface requirements required to implement the system in alignment with business expectations.

2. PURPOSE

The purpose of this BRD is to:

- Document all **functional and operational requirements** for the KiyaAI Credit Assessment System.
- Provide a detailed understanding of **screens, workflows, user roles, system processes, data flows, APIs, and infrastructure needs**.
- Establish a single source of truth for stakeholders including:
 - Product Owners
 - Business Analysts
 - Technical Architects
 - Developers
 - QA Teams
 - End Users
 - Implementation & Deployment Teams
- Define expected behaviors of each module to ensure:
 - Consistency in development
 - Clarity for UI/UX teams
 - Precision for backend/API development
 - Alignment with regulatory and business objectives
- Bridge the gap between business expectations and technical implementation by clearly articulating:
 - Use cases

- Flows
- Acceptance criteria
- Dependencies

The BRD ensures that all teams have a unified understanding of **what the system must deliver** and **how it should function**.

3. SCOPE OF THE BRD

The scope of this BRD includes documenting the full set of business and system requirements for the **design, development, deployment, and operation** of the KiyaAI Credit Assessment System.

3.1 In-Scope Functional Areas

This BRD covers detailed requirements for the following modules:

A. Core User Modules

1. **Login & Authentication**
2. **Dashboard & Analytics**
3. **Credit Assessment Module**
 - Entity details
 - Financial data capture (manual + OCR + file upload)
 - Ratio calculations
 - Business and qualitative scoring
 - Final rating computation
 - Maker-checker approvals (if applicable)
4. **Rating Templates Management**
5. **Parameters Management**
6. **Reports & Document Generation**

B. Administrative Modules

7. **Role Management**
8. **User Management**
9. **RBAC (Role-Based Access Control)**
10. **Master Data Management**
 - Country, State, City
 - Lookup management
11. **System Configurations**
 - Email settings

- Email templates
- Module management

C. Data Management Modules

12. Data Ingestion

- Financial data ingestion
- OCR processing
- Excel/CSV file processing
- Data validation
- Gaps identification

D. Technical & Deployment Scope

13. API Catalog

- Internal APIs
- Expected third-party integrations

14. Deployment & Infrastructure Requirements

- Hosting
- Database (Oracle)
- Software prerequisites

15. System Documents

- FSD
- User Manual
- Deployment Guide

3.2 Out-of-Scope Areas

The following items are outside the scope of this BRD unless otherwise requested:

- Enterprise LOS integration (Loan Origination System)
- Credit Bureau API integrations (unless added as per customer requirement)
- Mobile application development
- Advanced ML risk modeling
- Automated underwriting engine
- Third-party compliance validation tools

3.3 Stakeholders

- **Business Team** – Defines assessment rules, workflows, templates

- **IT Team** – Infrastructure, security, deployment
- **Development Team** – UI, backend, APIs
- **QA Team** – Testing and validations
- **Operations Users** – Credit analysts, approvers
- **Administrators** – Role, user and system configuration

3.4 Assumptions

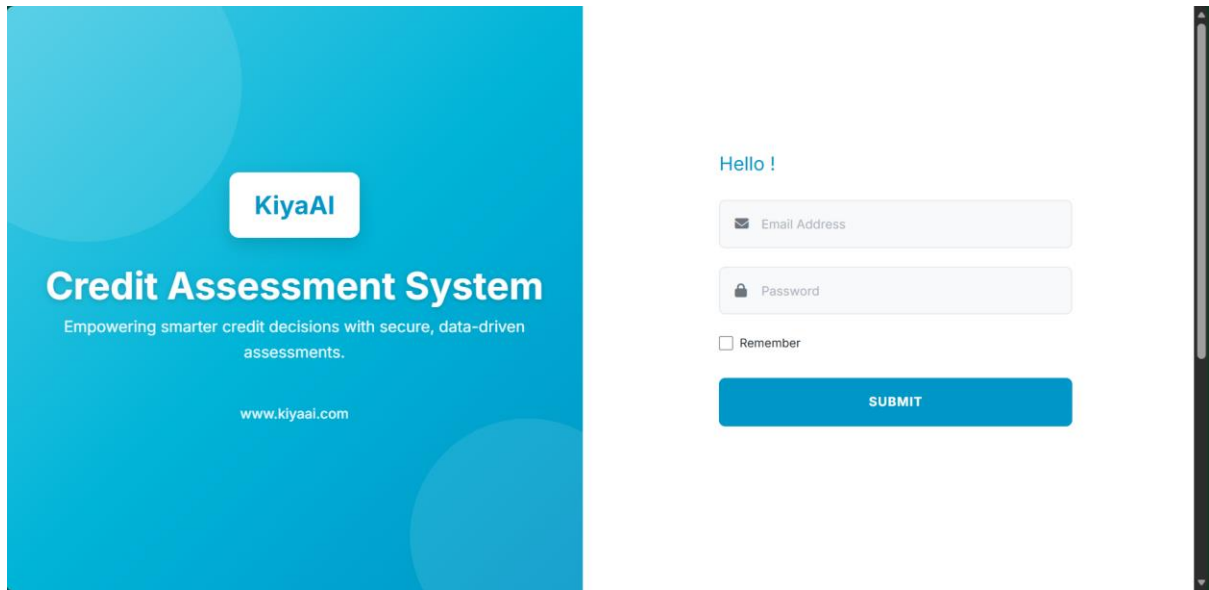
- The system will run on a secure network with HTTPS enabled.
- Oracle Database will be available as the primary data source.
- All users will be authenticated using username/password unless SSO is enabled in future.
- Master data will be provided prior to UAT.

3.5 Dependencies

- Availability of financial data input formats.
- Integration availability for OCR engine.
- Access control rules provided by business team.
- Email server configuration details.

4. Features & Functionalities

4.1. Login



The Login module provides secure entry into the KiyaAI Credit Assessment System. Users can authenticate either through **ADS (Active Directory Services)** or through **Manual Login** using email and password. The objective is to ensure only authorized users gain access to the system.

The Login screen contains:

- **Left Panel** → Welcome branding, logo, product title
- **Right Panel** → Login form
- **Fields:**
 - Email
 - Password
 - Remember Me
- **Buttons:**
 - Login
 - ADS Login (if enabled)
- **Error Display:** Invalid credential banner
- Fully responsive layout

Manual Login

- User enters **Email + Password**.
- System validates format and required fields.
- Authentication request sent to backend.
- On success → dashboard loads.

- On failure → error message shown with shake animation.

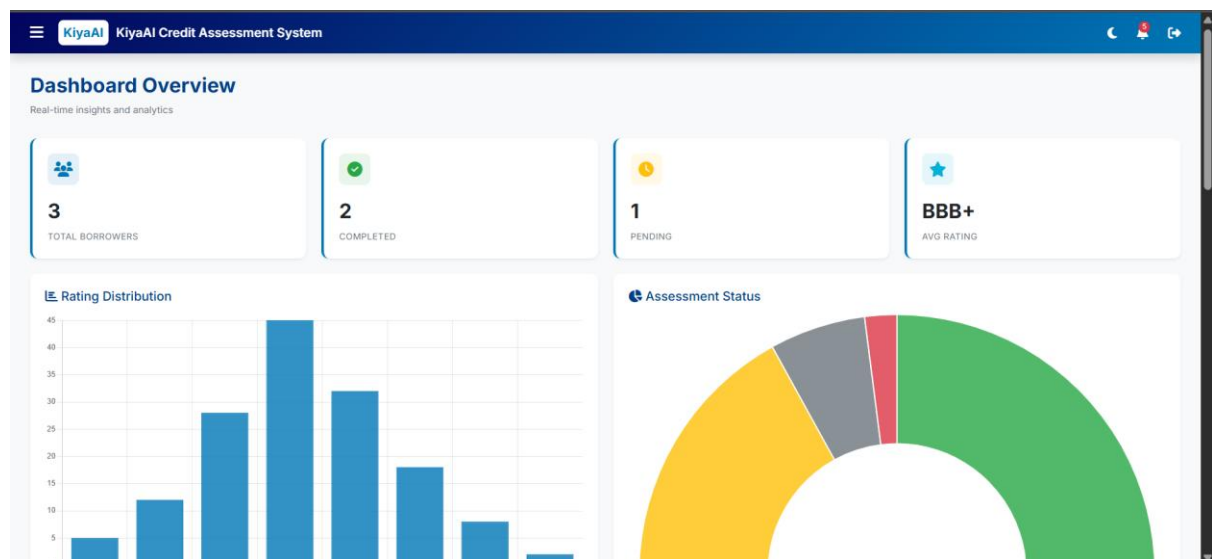
ADS Login

- User clicks **"Login with ADS"**.
- System redirects to configured ADS/SSO endpoint.
- ADS validates user credentials.
- On success → system receives user profile & roles.
- User is logged into the application without manual credential entry.

Security

- All credentials transmitted over HTTPS (If SSL Certificate implemented).
- Invalid attempts logged in audit trail.
- User session managed via token-based authentication.
- Logout clears token and session.

4.2. Dashboard



The Dashboard provides a real-time overview of credit assessment activities, borrower statistics, rating outcomes, and workflow status. It acts as the user's landing page after login and presents high-level operational insights.

The Dashboard contains the following key UI elements:

A. Summary Cards

- Total Borrowers

- Completed Assessments
- Pending Assessments
- Average Rating

B. Charts & Analytics

- Rating Distribution (Bar Chart)
- Assessment Status Distribution (Pie Chart)
- Monthly Assessment Trend (Line Chart)

C. Recent Assessments Table

- ID
- Entity Name
- Template
- Date
- Status
- Rating
- Quick Actions: Edit, Tag CIF, 360° View

A “New Assessment” button is also available to directly start the assessment workflow.

FUNCTIONAL DETAILS

Summary Statistics

- Fetches aggregated data from backend services.
- Auto-refresh on page load.
- Reflects live assessment counts and performance indicators.

Charts

- **Rating Distribution:** Shows number of entities per rating grade.
- **Assessment Status Chart:** Displays Pending / Completed / Draft statuses.
- **Trend Chart:** Monthly count of assessments over the selected period.

Charts help users understand portfolio risk patterns and workloads.

Recent Assessments

- Displays latest assessment records.
- Sortable and supports quick navigation to Edit/View.
- “New Assessment” initiates Step-1 of the Assessment Module.
- 360° customer view button opens consolidated customer profile (if implemented).

- Records show status badges (Draft, Pending, Completed).

User Actions

- Navigate to Assessment Module
- Continue editing previous assessments
- Tag CIF to assessments lacking customer linkage
- Trigger analytics or 360° views

Client Additions –

- Dashboard/Landing Page must show each case with:
 - Current workflow stage**
 - Status indicators
- Helps users track assessment stage quickly.

4.3. Rating Templates

KiyaAI Credit Assessment System						
Rating Templates						
Manage credit rating templates and configurations						
+ New Template						
Template ID	Template Name	Category	Industry	Parameters	Status	Actions
TMPL_001	Corporate Credit Rating	CORPORATE	undefined	0	ACTIVE	View Edit Delete
TMPL_002	SME Quick Assessment	SME	undefined	0	ACTIVE	View Edit Delete
TMPL_003	Individual Borrower Rating	INDIVIDUAL	undefined	0	ACTIVE	View Edit Delete
TMPL_CONTRACTING_001	Contracting & Services Rating	CONTRACTING & SERVICES	Construction, Engineering, Services	10	ACTIVE	View Edit Delete

Add New Template

Template Name *
e.g., Corporate Credit Rating Model

Category *
Select category

Industry Sector *
Select industry

Status *
Active

Created By
Admin User

Description
Enter template description and purpose...
Provide a brief description of this template's use case and target entities.

Note: After creating the template, you can add parameters from the Parameters Management section.

Cancel Save as Draft Save & Activate

Template ID	Template Name
TMPL_001	Corporate Credit Rating
TMPL_002	SME Quick Assessment
TMPL_003	Individual Borrower Rating
TMPL_CONTRACTING_001	Contracting & Services Rating

Parameters	Status	Actions
0	ACTIVE	[Icon] [Icon] [Icon]
0	ACTIVE	[Icon] [Icon] [Icon]
0	ACTIVE	[Icon] [Icon] [Icon]
10	ACTIVE	[Icon] [Icon] [Icon]

Edit Template

Template Name *
Corporate Credit Rating

Category *
Corporate

Industry Sector *
[Empty]

Status *
Active

Created By
undefined

Description
Enter template description and purpose...
Provide a brief description of this template's use case and target entities.

Note: After creating the template, you can add parameters from the Parameters Management section.

Cancel Save as Draft Save & Activate

Template ID	Template Name
TMPL_001	Corporate Credit Rating
TMPL_002	SME Quick Assessment
TMPL_003	Individual Borrower Rating
TMPL_CONTRACTING_001	Contracting & Services Rating

Parameters	Status	Actions
0	ACTIVE	[Icon] [Icon] [Icon]
0	ACTIVE	[Icon] [Icon] [Icon]
0	ACTIVE	[Icon] [Icon] [Icon]
10	ACTIVE	[Icon] [Icon] [Icon]

The Rating Templates module allows administrators and credit analysts to **create, configure, and manage rating templates** used during credit assessment. Each template defines the structure of parameters, categories, weights, and industry applicability.

The Templates screen contains:

A. Template List Table

Columns include:

- Template ID
- Template Name
- Category (Corporate, SME, Real Estate, etc.)
- Industry
- Parameter Count

- Status (Active / Draft / Inactive)
- Actions (Edit, Activate/Deactivate)

B. Header Actions

- **New Template** button opens the Add/Edit Template Modal.

C. Add/Edit Template Modal

Includes fields:

- Template Name
- Category
- Industry
- Status
- Created By
- Description
- Save as Draft / Save & Activate buttons

The modal also displays a note that parameters can be added from the Parameters module after template creation.

FUNCTIONAL DETAILS

Create Template

- User clicks **New Template**.
- Modal opens to input mandatory fields.
- On save:
 - If “Save as Draft” → Template saved as Draft.
 - If “Save & Activate” → Template becomes Active.
- System records:
 - Creator
 - Creation timestamp
 - Status

Templates cannot be used in assessments unless status = Active.

Edit Template

- Selecting “Edit” from the table opens the modal pre-filled.
- Editable fields:
 - Name
 - Category

- Industry
- Description
- Status
- Updating an Active template may require versioning (based on customer requirement).

Template Status Management

Statuses:

- **Active** → Available for assessments
- **Draft** → Editable, not usable
- **Inactive** → Archived, not selectable

Status changes will be logged in the audit trail.

Template-Parameter Linkage

- Templates define structure only.
- Parameters (weights, formulas, qualitative options) are configured in the **Parameters Module**.
- Each template may have multiple parameters grouped by categories.

Templates Modal's System will have, as per the client's shared data –

1. Contracting & Service Template
2. Trading & Manufacturing Template
3. Investment Real estate & finance
4. Personal Net Worth Declaration Borrowers Grading

Client Additions –

4.3.x Template Associations & Auto-Selection Enhancements

1. Multi-Select Category & Sector Mapping
 - Rating Templates must support:
 - Multi-select Categories
 - Multi-select Industry Sectors
 - Admin can assign a single template to multiple sectors & categories.
2. Auto-Selection of Rating Model
 - During assessment, the Rating Model/template must be automatically selected based on:

- Entity Type
- Industry Sector
- If user chooses a different template, system must display a warning prompt stating recommended template and allowing override.

4.4. Parameters Management

Parameters Management
Define and manage rating parameters for templates

[+ Add Parameter](#)

All Parameters Contracting & Services Rating

Parameter ID	Template	Section	Name	Weight	Type	Actions
PARAM1763015701391	Contracting & Services Rating	Financial Parameters	Current Ratio	10%	Calculated	Edit Delete
PARAM1763015828333	Contracting & Services Rating	Financial Parameters	Debt-to-Equity Ratio	10%	Calculated	Edit Delete
PARAM1763015912996	Contracting & Services Rating	Financial Parameters	Interest Coverage Ratio	10%	Calculated	Edit Delete
PARAM1763015969852	Contracting & Services Rating	Financial Parameters	Net Profit Margin	10%	Calculated	Edit Delete
PARAM1763016022874	Contracting & Services Rating	Financial Parameters	Cash Flow Adequacy	10%	Calculated	Edit Delete

Parameters Management
Define and manage rating parameters for templates

[+ Add Parameter](#)

Add New Parameter

Template *

Section *

Parameter Name *

Weight (%) *

Data Type

Score Type

Description

Formula (for auto-calculated parameters)
e.g., currentAssets / currentLiabilities
Use field names from Financial data

[Save Parameter](#) [Cancel](#)

Qualitative Scoring Options
Define dropdown options with their corresponding scores

Option Label (e.g., Excellent) Score (0-5) Description

[+ Add Option](#)

[Save Parameter](#) [Cancel](#)

The Parameters Management module allows administrators to **define, configure, and maintain scoring parameters** used inside rating templates.

Each parameter contributes to the overall credit score through weights, formulas, scoring methods, or qualitative options.

This module ensures consistency and flexibility in rating model configuration.

The screen consists of:

A. Parameter List Table

Columns:

- Parameter ID
- Template Name
- Section (Financial / Business / Risk)
- Parameter Name
- Weight %
- Type (Ratio / Percentage / Number / Qualitative / Score)
- Actions (Edit/Delete)

B. Header Action

- **Add Parameter** button → opens Parameter Form

C. Parameter Form (Add/Edit)

Fields include:

- Template
- Section
- Parameter Name
- Weight (%)
- Data Type
- Score Type (Auto / Manual)
- Description
- Formula (for auto-calculated parameters)
- **Qualitative Options Section** (dynamic fields)

Visibility:

- Qualitative section appears only when Data Type = *Qualitative*.

FUNCTIONAL DETAILS

Add Parameter

- User selects template & section.
- Enters parameter details including:
 - Name

- Weight
 - Type
 - Description
 - Formula (optional)
- If Data Type is **Qualitative**, user adds:
 - Options
 - Score for each option
- Save stores parameter under selected template.

Edit Parameter

- Opens the form with existing parameter data.
- Allows modification of:
 - Weight
 - Description
 - Formula
 - Options (if qualitative)
- Template link may be restricted for editing (based on configuration rules).

Parameter Types

- **Ratio** → Formula-based computation
- **Percentage** → Numeric input
- **Number** → Raw numeric values
- **Score (1-5)** → Manual or auto
- **Qualitative** → Dropdown with score mapping

Weight Validation

- Total weight across all parameters for a template must sum up to **100%**.
- System must:
 - Warn user on exceeding
 - Block save if invalid

Parameter-Template Relationship

- One template → Multiple parameters
- Parameters grouped by:
 - Financial Parameters
 - Business Parameters
 - Risk Parameters

Displayed in assessment steps accordingly.

Data Binding

- Parameters influence:
 - Ratio calculations
 - Business scoring
 - Final rating outcome

Values captured during assessment populate these parameters.

Parameters as per shared data

1. Contracting & Service Template

✓ 1. Customer Financial Information Table		
Sr. No	Parameter	Description
1	Revenue	Annual revenue of the customer
2	Cost of Goods Sold	Direct cost of goods sold
3	Net Profit	Profit after all expenses
4	Total Current Assets	Sum of short-term assets
5	Affiliates Due	Amount due from affiliates
6	Investment Assets	Financial investment assets
7	Total Assets	Total assets (current + non-current)
8	Total Current Liabilities	Current obligations
9	Total Debt	Total CL + Long-term liabilities
10	Total Bank Borrowings	Total outstanding bank loans
11	Total Liabilities incl. Owner's Funds	Overall liabilities
12	Net Worth	Shareholders' equity
13	Operating Cash	Operating cash flow

✓ 2. Non-Financial Information Table		

Sr. No	Parameter	Description
1	Constitution and Establishment	Type of entity & incorporation details
2	Number of Years in Business	Business age
3	Unutilized Credit Limits	Available unused credit with lenders
4	Financial Planning & Control	Risk management & internal audit practices
5	Position in the Market	Competitive standing
6	Govt./Private Contracting	Type of client portfolio
7	Sensitivity to Economic Shocks	Stability against external disruptions
8	Contractor/Subcontractor Grade	Classification/grading of contractor
9	Past Track Record	History of service delivery
10	Competition & Demand	Industry competition intensity
11	Qualified Resources & Management	Availability of skilled human and physical resources

✓ 3. Mandatory Adjustments Table

Parameter	Description
Age of Financial Information	Recency of financial statements
Quality of Financial Information	Reliability, audit quality
Historical Account Performance	Customer repayment history
Qualification of Audit Report	Whether audit report has qualifications

✓ 4. Discretionary Overrides Table (Complete Professional Version)

Category	Override Reason	Max Upgrade	Max Downgrade
GEN	Management changes	2	2
GEN	Change in Regulation	1	1
GEN	Length of relationship – Strong / Weak	1	1
GEN	External Relationship (Part of strong group / troubled group)	1	1
GEN	Deposit availability / Ancillary business	1	0
GEN	Bankruptcy, Liquidation	0	3
GEN	Deterioration of group accounts exposure	0	2
GEN	Unresolved conflict between owners	0	2
GEN	Death of a key person	0	2
GEN	Non-existence of succession plan	0	1
FIN	Income Diversification	1	1

BUSS. PERF	Strategic contract signed / loss of key contract	2	2
BUSS. PERF	Change in Market Position	2	2
BUSS. PERF	Performance vs. Budget (better/worse)	1	1
BUSS. PERF	Reputation (Strong / Poor)	1	1
LEGAL	Ongoing legal dispute / Court order	0	2
ECONOMIC	Drastic fall in capital markets	0	1
ECONOMIC	Drastic fall in oil prices	0	1
ECONOMIC	Global economic deterioration	0	2
POLITICAL	Social disturbances / unrest	0	1
ASSIGNMENT	100% Govt. sector contracts with legal assignment	2	0
ASSIGNMENT	Majority contracts (>75%) with strong assignment	1	0
ASSIGNMENT	>50% contracts without credible assignment	0	1
Account Conduct	Persistent ever-greening (>2 years)	0	1

Qualitative Factor – Relative Risk Ordering - Score

F1 – Constitution and Establishment

Range	Relative Risk Order	Score
Government / Semi-government	1	29
Public Shareholding Companies	2	22
Closed Shareholding Companies	3	15
Partnership with >50% ownership / Sole Proprietorship	4	7
Other Partnership or Establishment	5	0

F2 – Number of Years in Business

Range	Relative Risk Order	Score
More than 20 years	1	29
10 – 20 years	2	22
2 – 10 years	3	11
Less than 2 years	4	0

F3 – Availability of Unutilized Credit Lines

Range	Relative Risk Order	Score
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Available & utilized < 20%	1	29
Available & utilized 20–70%	2	22
Sole Bank	3	15
Available & utilized > 70%	4	7
Over-utilization beyond limits	5	0

F4 – Financial Planning & Internal Control

Range	Relative Risk Order	Score
Not Applicable	0	23
Good	1	27
Moderate	2	18
No financial planning & control	3	9

F5 – Position in the Market

Range	Relative Risk Order	Score
Market Leaders	1	29
Strong Position	2	22
Below Average	3	11
New Entrant	4	0

F6 – Govt. / Private Sector Contracting

Range	Relative Risk Order	Score
Not Applicable (Neutral Score)	0	23
100% Govt. / Public Sector	1	29
>75% Govt. / Public Sector	2	22
>50% Govt. + strong private	3	15
Majority private – net worth known	4	7
Majority private – net worth unknown	5	0

F7 – Sensitivity to Economic Shocks

Range	Relative Risk Order	Score
-------	---------------------	-------

Unlikely to be affected	1	29
Minor impact	2	22
Likely to see impact	3	11
Highly sensitive	4	0

F8 – Contractor / Subcontractor Grade

Range	Relative Risk Order	Score
Not Applicable (Service Companies)	0	23
Grade I	1	26
Grade II	2	22
Grade III	3	17
Grade IV / Ungraded	4	12

F9 – Past Track Record

Range	Relative Risk Order	Score
Not Applicable	0	23
Excellent	1	31
Good	2	21
Average	3	7
Poor	4	-8

F10 – Competition & Demand

Range	Relative Risk Order	Score
Low competition & high demand	1	29
High competition & high demand	2	22
Low competition & low demand	3	11
High competition & low demand	4	0

F11 – Resource & Management Capability

Range	Relative Risk Order	Score
Well Qualified	1	33

Partially Qualified	2	9
Not adequately qualified	3	-14

Quantitative Factors

✓ MASTER FINANCIAL RATIO SCORING TABLE

1. AffDueToNW (Affiliates Due to Net Worth Ratio)

Measures: *How much affiliates owe compared to company's net worth.*

Label	From	To	Points
AffDueToNW < 0.01	-9999	0.01	33
$0.01 \leq \text{AffDueToNW} < 0.1$	0.01	0.1	25.6
$0.1 \leq \text{AffDueToNW} < 0.2$	0.1	0.2	21
$0.2 \leq \text{AffDueToNW} < 0.66$	0.2	0.66	15.5
AffDueToNW ≥ 0.66	0.66	9999	13.6

2. Current Ratio

Formula: **Current Assets / Current Liabilities**

Higher = safer liquidity.

Label	From	To	Points
CurrentRatio < 0.78	0	0.78	7.1
$0.78 \leq \text{CurrentRatio} < 0.96$	0.78	0.96	9.5
$0.96 \leq \text{CurrentRatio} < 1.1$	0.96	1.1	11.9
$1.1 \leq \text{CurrentRatio} < 1.47$	1.1	1.47	15.7
$1.47 \leq \text{CurrentRatio} < 2.96$	1.47	2.96	27.1
$2.96 \leq \text{CurrentRatio} < 5.38$	2.96	5.38	34
CurrentRatio ≥ 5.38	5.38	9999	41.2

3. Debt Equity Ratio (DER)

Formula: **Total Debt / Net Worth**

Lower = safer.

Label	From	To	Points
DER < 0.16	-9999	0.16	41
$0.16 \leq \text{DER} < 0.42$	0.16	0.42	33.9

$0.42 \leq \text{DER} < 1.06$	0.42	1.06	21.5
$1.06 \leq \text{DER} < 1.48$	1.06	1.48	18.9
$1.48 \leq \text{DER} < 1.93$	1.48	1.93	14.5
$1.93 \leq \text{DER} < 4$	1.93	4	10
$\text{DER} \geq 4$	4	9999	7.4

4. OpsRev/ExtLiab (Operating Revenue / External Liabilities)

Higher = better repayment capacity.

Label	From	To	Points
< -1.6	-9999	-1.6	1.8
$-1.6 \leq x < -0.4$	-1.6	-0.4	6.8
$-0.4 \leq x < -0.15$	-0.4	-0.15	11.8
$-0.15 \leq x < -0.03$	-0.15	-0.03	16.8
$-0.03 \leq x < 0.03$	-0.03	0.03	21.8
$0.03 \leq x < 0.13$	0.03	0.13	26.8
$0.13 \leq x < 0.45$	0.13	0.45	31.8
$0.45 \leq x < 2.1$	0.45	2.1	36.8
≥ 2.1	2.1	9999	41.8

5. InvAssetsToTA (Investment Assets to Total Assets)

Label	From	To	Points
< 0.09	0.01	0.09	31
$0.09 \leq x < 0.24$	0.09	0.24	19.1
$0.24 \leq x < 0.64$	0.24	0.64	10
≥ 0.64	0.64	1.5	6.7

6. Net Profit Margin

Label	From	To	Points
< 0	-9999	0	14.9
$0 \leq x < 0.06$	0	0.06	18.8
$0.06 \leq x < 0.21$	0.06	0.21	26.9

≥ 0.21	0.21	9999	35.2
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7. WCT (Working Capital Turnover / Working Capital to Turnover)

Higher = better working capital health.

Label	From	To	Points
< -0.48	-9999	-0.48	11.8
$-0.48 \leq x < 0$	-0.48	0	14.4
$0 \leq x < 0.41$	0	0.41	17
$0.41 \leq x < 1.2$	0.41	1.2	23.1
$1.2 \leq x < 1.88$	1.2	1.88	27.5
$1.88 \leq x < 4.52$	1.88	4.52	31.8
≥ 4.52	4.52	9999	33.6

8. d_DebtRatio (Change in Debt Ratio)

Measures: how much debt increased or decreased YoY.

Label	From	To	Points
< -0.2	-9999	-0.2	40.4
$-0.2 \leq x < -0.11$	-0.2	-0.11	31.3
$-0.11 \leq x < 0.33$	-0.11	0.33	20.5
≥ 0.33	0.33	9999	12.3

2. Investment Real estate & finance Template

☒ 1. FINANCIAL INFORMATION TABLE

Sr. No	Parameter	Description
1	Balance Sheet Info	Summary of assets, liabilities & equity
2	Gross Profit	Revenue - COGS
3	General & Administrative Expenses	Non-operating and administrative costs
4	Interest	Interest expense on borrowings
5	Net Profit	Profit after all expenses
6	Inventory	Total inventory held

7	Total Current Assets	Short-term assets
8	Total Assets	All assets combined
9	Total Current Liabilities	Short-term liabilities
10	Total Long Term Liabilities	Long-term debt obligations
11	Current Debt	Short-term debt obligations
12	Total Debt	Current CL + LT Liabilities
13	Total Liabilities incl. Owner's Funds	Overall obligations including capital
14	Total Bank Borrowings	Total outstanding from banks
15	Unquoted Asset	Non-listed investments
16	Unrealized Gains	Gains not yet realized
17	Operating Cash	Cash generated from operations
18	Net Worth	Shareholders' equity

2. NON-FINANCIAL INFORMATION TABLE

Sr. No	Parameter	Description
1	Constitution & Establishment	Business type & incorporation
2	Management Character / Style	Leadership credibility & governance
3	Number of Years in Business	Tenure & experience
4	Availability of Unutilized Credit Lines	Unused credit from other lenders
5	Financial Planning & Control	Internal audit & risk management quality
6	Asset Quality – Investment Diversification	Diversification of investment assets
7	Asset Quality – Investment Risk	Risk profile of investment assets
8	Asset Quality – Real Estate Concentration	Concentration & type of real estate assets

3. MANDATORY ADJUSTMENTS TABLE

Sr. No	Parameter	Description
1	Age of Financial Information	Recency of financial statements
2	Quality of Financial Information	Reliability & audit quality
3	Historical Account Performance	Past repayment history
4	Qualification of Audit Report	Nature of audit qualifications

✓ 4. DISCRETIONARY OVERRIDES TABLE (DETAILED & PROFESSIONAL)

Class	Override Reason	Max Upgrade	Max Downgrade
GEN	Management changes	2	2
GEN	Change in Regulation	1	1
GEN	Length of relationship – strong/weak	1	1
GEN	External relationship – strong group / troubled group	1	1
GEN	Deposit availability / ancillary business	1	0
GEN	Bankruptcy, Liquidation	0	3
GEN	Deterioration of other group exposures	0	2
GEN	Unresolved conflict between owners	0	2
GEN	Death of a key person	0	2
GEN	No succession plan	0	1
FIN	Income diversification	1	1
BUSS. PERF	Strategic contract signed / loss of contract	2	2
BUSS. PERF	Change in market position	2	2
BUSS. PERF	Performance vs budget (better/worse)	1	1
BUSS. PERF	Reputation (Good/Poor)	1	1
LEGAL	Ongoing legal dispute / court order	0	2
ECONOMIC	Drastic fall in capital markets	0	1
ECONOMIC	Drastic fall in oil prices	0	1
ECONOMIC	Global economic deterioration	0	2
POLITICAL	Social disturbances / uprisings	0	1
ACCOUNT CONDUCT	Persistent evergreening (>2 years)	0	1

Qualitative Factor – Relative Risk Ordering - Score

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F1	Constitution and Establishment	Government / Semi-government	1	50
		Public Shareholding Companies	2	38

		Closed Shareholding Companies	3	25
		Partnership (with >50% owner guarantee) / Sole Proprietorship	4	13
		Other Partnership or Establishment	5	0
F2	Management Character / Style	Risk Conservative Management Approach	1	50
		Moderate & Controlled Risk Taking	2	38
		High Risk Taker (with analysis & provisions)	3	19
		Very High Risk Taker (no analysis)	4	0
F3	Number of Years in Business	More than 20 years	1	50
		10-20 years	2	38
		2-10 years	3	19
		Less than 2 years	4	0
F4	Availability of Unutilized Credit Lines	Lines available, utilization <20%	1	50
		Lines available, utilization 20-70%	2	38
		Sole Bank	3	25
		Lines available, utilization >70%	4	13
		Utilization beyond limits	5	0
F5	Existence of Financial Planning & Control	Not applicable	0	16
		Good	1	50
		Moderate	2	25
		No financial planning & control	3	0
F6	Assets Quality - Investment (Diversification)	N/A (Only for Real Estate Companies)	0	16
		Well diversified across sectors & products	1	50
		Minor concentration (Sector/Product)	2	38
		High concentration in multiple sectors/products	3	19
		Very high concentration / No information	4	0
F7	Assets Quality - Investment (Asset's Risk)	N/A (Only Real Estate)	0	16
		High quality fixed income + minor diversified equity	1	50
		Well balanced, mostly listed, low volatility	2	38
		Risky investments, equity heavy	3	19

		Highest risk assets / non-listed / no information	4	0
F8	Assets Quality – Real Estate	N/A (Only for Investment Companies)	0	16
		No diversified RE portfolio	1	50
		Some concentration in risky properties (office)	2	38
		Significant concentration in office properties	3	19
		Very high concentration / No information	4	0

Quantitative Factors

✓ 1. Current Debt to Total Debt Ratio

Label	From	To	Points
CurrentDebtToTotalDebt < 0.4	0	0.4	24.7
$0.4 \leq \text{CurrentDebtToTotalDebt} < 0.79$	0.4	0.79	21.9
$0.79 \leq \text{CurrentDebtToTotalDebt} < 0.92$	0.79	0.92	19.2
CurrentDebtToTotalDebt ≥ 0.92	0.92	1	10.9

✓ 2. Debt Equity Ratio (DER)

Label	From	To	Points
DER < 0.17	-999	0.17	41.4
$0.17 \leq \text{DER} < 0.38$	0.17	0.38	33.9
$0.38 \leq \text{DER} < 1.02$	0.38	1.02	29
$1.02 \leq \text{DER} < 1.5$	1.02	1.5	18.8
$1.5 \leq \text{DER} < 2.5$	1.5	2.5	11.4
DER ≥ 2.5	2.5	999	3.9

✓ 3. Interest Coverage Ratio (ICR)

Label	From	To	Points
ICR < 0	-999	0	12.2
$0 \leq \text{ICR} < 1$	0	1	14
$1 \leq \text{ICR} < 1.96$	1	1.96	16.4
$1.96 \leq \text{ICR} < 3.3$	1.96	3.3	18.1

$3.3 \leq \text{ICR} < 5.9$	3.3	5.9	22
$5.9 \leq \text{ICR} < 10.1$	5.9	10.1	24.6
$\text{ICR} \geq 10.1$	10.1	999	26.3

✓ 4. ROA (Return on Assets)

Label	From	To	Points
$\text{ROA} < -0.09$	-9999	-0.09	-4.5
$-0.09 \leq \text{ROA} < 0.01$	-0.09	0.01	-0.7
$0.01 \leq \text{ROA} < 0.04$	0.01	0.04	9.4
$0.04 \leq \text{ROA} < 0.07$	0.04	0.07	17.1
$0.07 \leq \text{ROA} < 0.13$	0.07	0.13	23.4
$0.13 \leq \text{ROA} < 0.23$	0.13	0.23	40.9
$\text{ROA} \geq 0.23$	0.23	999	59

✓ 5. Change in Debt Equity Ratio ($\Delta \text{DER} / d2_DER$)

Label	From	To	Points
$d2_DER < -0.6$	-999	-0.6	18.8
$-0.6 \leq d2_DER < 0$	-0.6	0	8
$0 \leq d2_DER < 1.03$	0	1.03	-5
$d2_DER \geq 1.03$	1.03	999	-11

✓ 6. Change in Quick Ratio ($d2_QuickRatio$)

Label	From	To	Points
$d2_QuickRatio < -1.04$	-999	-1.04	0
$-1.04 \leq d2_QuickRatio < -0.73$	-1.04	-0.73	10
$-0.73 \leq d2_QuickRatio < 0.21$	-0.73	0.21	25
$d2_QuickRatio \geq 0.21$	0.21	999	40

✓ 7. Change in ROA ($d2_ROA$)

Label	From	To	Points
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$d2_ROA < -0.22$	-9999	-0.22	-10.5
$-0.22 \leq d2_ROA < -0.07$	-0.22	-0.07	5.9
$-0.07 \leq d2_ROA < 0$	-0.07	0	13.8
$0 \leq d2_ROA < 0.03$	0	0.03	17.9
$0.03 \leq d2_ROA < 0.08$	0.03	0.08	23.4
$0.08 \leq d2_ROA < 0.11$	0.08	0.11	33.9
$d2_ROA \geq 0.11$	0.11	999	58.9

✓ 8. Unrealized Gains to Net Profit

Label	From	To	Points
Unrealized Losses (<0)	-999	0	4
0	0	0.01	30
0 - 0.3	0.01	0.3	25
0.3 - 0.5	0.3	0.5	20
0.5 - 2	0.5	999	15

✓ 9. Unquoted Assets to Total Assets

Label	From	To	Points
0	0	0	30
0% - 5%	0.0000001	0.05	25
5% - 20%	0.05	0.2	18
20% - 40%	0.2	0.4	12
40% - 70%	0.4	0.7	6
$\geq 70\%$	0.7	99999	0

✓ 10. Operating Cash / External Liabilities

Label	From	To	Points
< -1.6	-99999999	-1.6	-4.1
-1.6 to -0.4	-1.6	-0.4	-0.1
-0.4 to -0.15	-0.4	-0.15	3.9
-0.15 to 0	-0.15	0	7.9

0 - 0.5	0	0.5	11.9
0.05 - 0.15	0.5	0.15	15.9
0.15 - 0.5	0.15	0.5	19.9
0.5 - 2	0.5	2	23.9
≥ 2	2	9999999	27.9

3. Personal Network Declaration Borrowers Grading

☒ 1. CUSTOMER FINANCIAL INFORMATION TABLE

Sr. No	Parameter	Description
1	Cash & Bank Balances	Total liquid cash and bank deposits
2	Total Current Assets	All short-term assets expected to be converted to cash within 12 months
3	Total Assets	Sum of current + long-term assets
4	Total Current Liabilities	Short-term liabilities due within 12 months
5	Total Long-Term Liabilities	Long-term financial obligations
6	Total Liabilities	Current + long-term liabilities
7	Net Worth	Shareholders' equity / owners' funds
8	Turnover	Total revenue / sales
9	Cost of Sales	Direct cost associated with generating sales
10	Gross Profit	Turnover - Cost of Sales
11	General & Administrative Expenses	Overheads and administrative expenses
12	Interest	Borrowing costs
13	Net Profit	Profit after all expenses

☒ 2. NON-FINANCIAL INFORMATION TABLE

Sr. No	Parameter	Description
1	Number of Years in Business	Duration of business operations
2	Existence of Legal Heirs	Succession planning existence & readiness

3	Management Character & Style	Governance quality and managerial behavior
4	Availability of Additional Financial Resources	Ability to raise funds during stress
5	Income Diversification	Diversity of income streams
6	Years in Relationship	Length of banking relationship
7	Accessibility	Ease of communication & collaboration
8	Credit History (Track Record)	Past repayment performance
9	Unquoted Assets	Non-listed / illiquid asset holdings

Qualitative Factor – Relative Risk Ordering - Score

✓ MASTER TABLE – COMPANY SPECIFIC & QUALITATIVE FACTORS

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F1	Number of Years in Business	More than 10 years	1	100
		6 – 10 Years	2	75
		2 – 5 Years	3	50
		New to industry (< 2 years), but relevant experience from other fields	4	25
		No or very limited relevant experience	5	0
F2	Existence of Legal Heirs	Client's age < 60 or clear succession plan with no legal risks	1	100
		Client's age ≥ 60 and unclear succession plan, possible legal risks	2	50
		Client's age ≥ 60 and no succession plan → high legal succession risk	3	0
F3	Management Character & Style	Cautious management with conservative risk appetite	1	100
		Balanced risk-taking and moderate risk appetite	2	75
		Aggressive but with high risk appetite	3	25
		Uncontrolled risk-taking (no analysis) and very high-risk attitude	4	0

F4	Availability of Additional Financial Resources	Credit lines available across lenders (<20% utilized)	1	100
		Credit lines available (20–70% utilized) OR CBK sole lender but <20% utilization	2	75
		Credit lines available but >70% utilized OR CBK sole lender (20–70% utilized)	3	50
		CBK sole lender; only small portion of line remains unutilized (>70% used)	4	25
		Client exceeded limits with CBK or other lenders	5	0
F5	Income Diversification	Strongly diversified across countries/regions/sectors	1	100
		Diversified across multiple countries or sectors	2	75
		Slight diversification but heavy dependency on one industry	3	37
		Concentrated in one sector and one region	4	0
F6	Years in Relationship	> 5 years	1	100
		3 – 5 years	2	75
		2 – 3 years	3	50
		1 – 2 years	4	25
		New to bank (≤ 1 year)	5	0
F7	Accessibility	Professional and accessible to RM	1	100
		Business manager is accessible	2	75
		Inaccessible temporarily (travel/lack of backup)	3	50
		Infrequent contact with RM		25
		Difficult to access / avoids bank	4	0
F8	Credit History (Track Record)	No excess drawings, delays, or breaches in 3 years	1	100
		Occasional breaches but resolved within one month	2	50
		New to bank (≤ 1 year), insufficient history	3	25

		Repeatedly late payments in last 3 years	4	0
F9	Unquoted Assets	<25% unquoted assets	1	100
		25–50% unquoted assets	2	67
		50–75% unquoted assets	3	33
		75–100% unquoted assets	4	0

Quantitative Factors

✓ 1. Debt Equity Ratio (DE)

Label	From	To	Points
DE < 8%	0	0.08	100
8% ≤ DE < 16%	0.08	0.16	75
16% ≤ DE < 29%	0.16	0.29	50
29% ≤ DE < 59%	0.29	0.59	25
DE ≥ 59%	0.59	999999999	0

✓ 2. Cash / Current Liabilities (Cash/CL)

Label	From	To	Points
Cash/CL < 0.0047	0	0.004688566	0
0.0047 ≤ Cash/CL < 0.025	0.004688566	0.025	25
0.025 ≤ Cash/CL < 0.100775194	0.025	0.100775194	50
0.100775194 ≤ Cash/CL < 0.638226402	0.100775194	0.638226402	75
Cash/CL ≥ 0.638226402	0.638226402	999999999	100

✓ 3. Interest Coverage Ratio (ICR)

Label	From	To	Points
ICR < 1.48	-99999999999	1.48159344	0
1.48 ≤ ICR < 2.97	1.48159344	2.972687772	25
2.97 ≤ ICR < 5.66	2.972687772	5.658536585	50
5.66 ≤ ICR < 11.63	5.658536585	11.625	75
ICR ≥ 11.63	11.625	999999999	100

✓ 4. EBIT / Total Current Liabilities (EBIT/TCL)

Label	From	To	Points
EBIT/TCL < 0.09	-9999999999	0.091104909	0
$0.09 \leq \text{EBIT/TCL} < 0.18$	0.091104909	0.18245093	25
$0.18 \leq \text{EBIT/TCL} < 0.39$	0.18245093	0.390353634	50
$0.39 \leq \text{EBIT/TCL} < 0.60$	0.390353634	0.603745918	75
EBIT/TCL ≥ 0.60	0.603745918	999999999	100

✓ 5. LN(Networth)

(Natural log of Net Worth)

Label	From	To	Points
LN(Networth) < 7.12	-99999999	7.116471779	0
$7.12 \leq \text{LN(Networth)} < 8$	7.116471779	8.003664679	25
$8 \leq \text{LN(Networth)} < 8.98$	8.003664679	8.982886178	50
$8.98 \leq \text{LN(Networth)} < 10.12$	8.982886178	10.1184078	75
LN(Networth) ≥ 10.12	10.1184078	9999999999	100

✓ 6. Growth in Net Worth (Δ Net Worth)

Label	From	To	Points
d_NW < -0.11	-99999999	-0.112212741	0
$-0.11 \leq d_NW < -0.006$	-0.112212741	-0.006246189	25
$-0.006 \leq d_NW < 0.064$	-0.006246189	0.064787206	50
$0.064 \leq d_NW < 0.16$	0.064787206	0.162455664	75
d_NW ≥ 0.16	0.162455664	9999999999	100

4. Trading & Manufacturing Template

✓ 1. FINANCIAL INFORMATION TABLE

Sr. No	Parameter	Description
1	Revenue	Total sales/turnover for the period

2	Cost of Goods Sold	Direct cost of producing goods/services sold
3	Net Profit	Profit after all expenses and taxes
4	Total Current Assets	Liquid and short-term assets realizable within a year
5	Affiliates Due	Receivables owed by group or affiliate entities
6	Investment Assets	Investments and financial holdings
7	Total Assets	Total current + non-current assets
8	Total Current Liabilities	Obligations due within 12 months
9	Total Debt (CL + LT Liabilities)	Combined short-term and long-term debt
10	Total Bank Borrowings	All outstanding bank borrowings
11	Total Liabilities incl. Owner's Funds	Total liabilities + shareholders' funds
12	Net Worth	Shareholders' equity
13	Operating Cash	Operating cash flow generated by business

2. NON-FINANCIAL INFORMATION TABLE

Sr. No	Parameter	Description
1	Constitution and Establishment	Business type, legal status, and founding details
2	Number of Years in Business	Duration of operations indicating stability
3	Unutilized Credit Limits	Available yet unused credit lines from other banks
4	Financial Planning & Control	Quality of internal audit and risk management
5	Position in the Market	Competitive strength & market presence
6	Competition / Supply & Demand	Market competitiveness and demand cycles
7	Sensitivity to Economic Shocks	Stability under economic disruptions
8	Customer Base	Spread, quality, and dependability of customers
9	Type of Debtors / Quality of Receivables	Risk level & collectability of receivables
10	Distribution Channel	Strength of sales/distribution infrastructure
11	Price Volatility Management	Ability to manage fluctuating commodity/market prices

3. MANDATORY ADJUSTMENTS TABLE

Sr. No	Parameter	Description
1	Age of Financial Information	How recent and updated financials are
2	Quality of Financial Information	Accuracy, reliability, and audit grade
3	Historical Account Performance	Past repayment & account behavior
4	Qualification of Audit Report	Whether auditor has raised any qualifications

✓ 4. DISCRETIONARY OVER-RIDES TABLE

Class	Override Reason	Max Upgrade	Max Downgrade
GEN	Management changes	2	2
GEN	Change in Regulation	1	1
GEN	Length of Relationship – Strong/Weak	1	1
GEN	External Relationship (Strong group / troubled group)	1	1
GEN	Deposit availability / Ancillary business	1	0
GEN	Bankruptcy, Liquidation	0	3
GEN	Deterioration of other group exposures	0	2
GEN	Unresolved conflict between owners	0	2
GEN	Death of a key person	0	2
GEN	Non-existence of succession plan	0	1
FIN	Income Diversification	1	1
BUSS. PERF	Significant contract signed / loss of major contract	2	2
BUSS. PERF	Change in Market Position	2	2
BUSS. PERF	Performance vs Budget (Better/Worse)	1	1
BUSS. PERF	Reputation (Strong/Poor)	1	1
LEGAL	Ongoing legal dispute / court order	0	2
ECONOMIC	Drastic fall in capital markets	0	1
ECONOMIC	Drastic fall in oil prices	0	1
ECONOMIC	Global economic deterioration	0	2
POLITICAL	Social disturbances / uprisings	0	1

Qualitative Factor – Relative Risk Ordering - Score

✓ MASTER TABLE – COMPANY SPECIFIC, BUSINESS SPECIFIC & SECTOR-SPECIFIC FACTORS

F1 – Constitution and Establishment

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F1	Constitution & Establishment	Government / Semi-government	1	29
		Public Shareholding Companies	2	22
		Closed Shareholding Companies	3	15
		Partnership with >50% personal guarantee / Sole Proprietorship	4	7
		Other Partnership or Establishment	5	0

F2 – Number of Years in Business

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F2	Number of Years in Business	More than 20 years	1	29
		10–20 years	2	22
		2–10 years	3	11
		Less than 2 years	4	0

F3 – Availability of Unutilized Credit Lines

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F3	Unutilized Credit Limits / Lines	Lines available, utilization <20%	1	27
		Lines available, utilization 20–70%	2	22
		Sole bank	3	18
		Lines available, utilization >70%	4	14
		Utilization exceeded limits	5	9

F4 – Financial Planning & Internal Controls

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F4	Financial Planning & Control	Not applicable	0	23
		Good	1	28
		Moderate	2	16
		No planning/control	3	5

F5 – Position in the Market

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F5	Position in the Market	Market Leaders	1	29
		Strong position among peers	2	22
		Average position	3	11
		New entrants	4	0

F6 – Competition / Supply & Demand

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F6	Competition / Supply & Demand	Low competition & high demand	1	28
		High competition & high demand	2	22
		Low competition & low demand	3	13
		High competition & low demand	4	5

F7 – Sensitivity to Economic Shocks

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F7	Economic Sensitivity	Unlikely to be affected	1	29
		Minor impact possible	2	22
		Likely to see impact	3	11
		Highly sensitive	4	0

F8 – Customer Base

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F8	Customer Base	Well-diversified and stable	1	29
		Good customer base above average	2	22
		Small & stable	3	11
		Very small & unstable	4	0

F9 – Quality of Receivables (Debtor Type)

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F9	Debtor Quality	Government / Semi-government entities	1	29
		Large Corporates	2	22
		SMEs	3	11
		Individuals	4	0

F10 – Distribution Channel

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F10	Distribution Channel	Effective	1	26
		Limited	2	12

F11 – Price Volatility Management

Factor ID	Factor Name	Factor Details	Relative Risk Ordering	Score
F11	Price Volatility Management	Not applicable	0	23
		Yes	1	29
		No	2	0

Quantitative Factors

☒ MASTER FINANCIAL RATIO SCORING TABLE

1. Affiliates Due to Net Worth (AffDueToNW)

Label	From	To	Points
$\text{AffDueToNW} < 0.01$	-9999	0.01	33
$0.01 \leq \text{AffDueToNW} < 0.1$	0.01	0.1	25.6
$0.1 \leq \text{AffDueToNW} < 0.2$	0.1	0.2	21
$0.2 \leq \text{AffDueToNW} < 0.66$	0.2	0.66	15.5
$\text{AffDueToNW} \geq 0.66$	0.66	9999	13.6

2. Current Ratio

Label	From	To	Points
$\text{CurrentRatio} < 0.78$	0	0.78	7.1
$0.78 \leq \text{CurrentRatio} < 0.96$	0.78	0.96	9.5
$0.96 \leq \text{CurrentRatio} < 1.1$	0.96	1.1	11.9
$1.1 \leq \text{CurrentRatio} < 1.47$	1.1	1.47	15.7
$1.47 \leq \text{CurrentRatio} < 2.96$	1.47	2.96	27.1
$2.96 \leq \text{CurrentRatio} < 5.38$	2.96	5.38	34
$\text{CurrentRatio} \geq 5.38$	5.38	9999	41.2

3. Debt Equity Ratio

Label	From	To	Points
$\text{DER} < 0.16$	-9999	0.16	41
$0.16 \leq \text{DER} < 0.42$	0.16	0.42	33.9
$0.42 \leq \text{DER} < 1.06$	0.42	1.06	21.5
$1.06 \leq \text{DER} < 1.48$	1.06	1.48	18.9
$1.48 \leq \text{DER} < 1.93$	1.48	1.93	14.5
$1.93 \leq \text{DER} < 4$	1.93	4	10
$\text{DER} \geq 4$	4	9999	7.4

4. Operating Revenue to External Liabilities (OpsRev/ExtLiab)

Label	From	To	Points
< -1.6	-9999	-1.6	1.8

$-1.6 \leq \text{Ratio} < -0.4$	-1.6	-0.4	6.8
$-0.4 \leq \text{Ratio} < -0.15$	-0.4	-0.15	11.8
$-0.15 \leq \text{Ratio} < -0.03$	-0.15	-0.03	16.8
$-0.03 \leq \text{Ratio} < 0.03$	-0.03	0.03	21.8
$0.03 \leq \text{Ratio} < 0.13$	0.03	0.13	26.8
$0.13 \leq \text{Ratio} < 0.45$	0.13	0.45	31.8
$0.45 \leq \text{Ratio} < 2.1$	0.45	2.1	36.8
≥ 2.1	2.1	9999	41.8

5. Investment Assets to Total Assets (InvAssetsToTA)

Label	From	To	Points
< 0.09	0.01	0.09	31
$0.09 \leq \text{Ratio} < 0.24$	0.09	0.24	19.1
$0.24 \leq \text{Ratio} < 0.64$	0.24	0.64	10
≥ 0.64	0.64	1.5	6.7

6. Net Profit Margin (NPM)

Label	From	To	Points
$\text{NPM} < 0$	-9999	0	14.9
$0 \leq \text{NPM} < 0.06$	0	0.06	18.8
$0.06 \leq \text{NPM} < 0.21$	0.06	0.21	26.9
≥ 0.21	0.21	9999	35.2

7. Working Capital Turnover (WCT)

Label	From	To	Points
$\text{WCT} < -0.48$	-9999	-0.48	11.8
$-0.48 \leq \text{WCT} < 0$	-0.48	0	14.4
$0 \leq \text{WCT} < 0.41$	0	0.41	17
$0.41 \leq \text{WCT} < 1.2$	0.41	1.2	23.1
$1.2 \leq \text{WCT} < 1.88$	1.2	1.88	27.5
$1.88 \leq \text{WCT} < 4.52$	1.88	4.52	31.8

WCT $\geq$ 4.52	4.52	999	33.6
-----------------	------	-----	------

8. Change in Debt Ratio (Δ Debt Ratio)

Label	From	To	Points
d_DebtRatio < -0.2	-9999	-0.2	40.4
-0.2 $\leq$ d_DebtRatio < -0.11	-0.2	-0.11	31.3
-0.11 $\leq$ d_DebtRatio < 0.33	-0.11	0.33	20.5
d_DebtRatio $\geq$ 0.33	0.33	9999	12.3

Reference –



RE_Credit Risk Rating Engine - Commercial Bank of Kuwait Requirements Gathering Sessions (Onsite).zip

4.5. Credit Assessment

The Credit Assessment module enables analysts to evaluate borrowers through a structured, step-by-step assessment workflow.

It covers entity verification, financial data capture, automated ratio computation, qualitative scoring, and generation of a final rating outcome.

This is the **core functional module** of the system.

Step 1 – Entity Details

Fields:

- Entity Name
- CIF Search & Deduplication
- Entity Type
- Industry Sector
- Rating Template
- Financial Year
- Registration No., Tax ID
- CIF Linking Options (Auto / Manual / Tag Later)

User enters entity information.

Template selection determines parameters in future steps.

CIF Search:

- User enters CIF → system checks duplicates.
- If match found → display match details with options:
 - “Link to Customer”
 - “Continue as New Customer”

“Tag CIF Later” allowed for new customers.

Client Additions –

1. Auto-Fetch Customer Information via CIF
 - When CIF is entered, system must auto-fetch:

- Entity Type
- Industry Sector
- Loaded data must appear pre-filled and editable.
- 2. **Business Activity Field**
 - Add a new field **Business Activity** to support customers without CIF.
- 3. **CINET / Ci-Net Number**
 - Add input field: **CINET Number**
 - Used for:
 - Validation of assessment
 - Future EOD file ingestion for bureau data
- 4. **Historical Auto-Population of Existing Customer Data**
 - For returning customers, system must auto-load all previously captured:
 - Entity information
 - Financial statements
 - Qualitative answers
 - Past rating results
 - Pre-filled values must remain editable.

The screenshot displays the 'Financial Statement Data' entry form within the 'KiyaAI Credit Assessment System'. The form is titled 'Enter amounts in thousands' and includes a 'Balance Sheet' section. It features two upload options: 'OCR Upload (Image/PDF)' and 'Excel/CSV Upload', both with 'Choose File' buttons and 'No file chosen' status. Below the uploads, the form is organized into three columns of input fields for the Balance Sheet:

- Total Assets ***: 100000000
- Current Assets ***: 200000
- Fixed Assets**: 1200000000
- Total Liabilities ***: 100000
- Current Liabilities ***: 100000
- Total Debt ***: 200000000
- Equity ***: 200000000
- Inventory**: 100000000
- Cash**: 100000000

Below the Balance Sheet, the 'Income Statement' section includes:

- Revenue ***: 100000000
- EBIT ***: 100000000
- Net Income ***: 100000000
- Interest Expense ***: 120000000
- Operating Cash Flow**: 200000000

At the bottom, there are navigation buttons: 'Previous', 'Save Draft', and 'Calculate & Continue ->'.

Step 2 – Financial Data

Sections:

- Balance Sheet

- Profit & Loss
- Cash
- Additional features:
- OCR Import (PDF/Image)
- Excel/CSV Upload
- Editable pre-filled fields after extraction

Flow

A. OCR Upload

- User uploads PDF/Image.
- OCR extracts values like:
 - Total Assets, Liabilities, Revenue, Expenses, etc.
- Populate fields for editing.

B. Excel/CSV Upload

- Predefined format.
- Auto maps financial data into fields.

C. Manual Entry

- All fields editable in thousands format.
- Mandatory fields validated before moving to next step.

Client Additions –

Support Full Financial Statement Structure

- UI must support complete statements:
 - Balance Sheet
 - Income Statement
 - Cash Flow
 - Financial Ratios
 - Projections Sheet

Alignment With Bank's Excel Template

- All rows/columns must match the bank's format exactly.

Excel Upload / OCR Extraction Enhancements

- Excel/OCR must:
 - Identify year columns
 - Identify financial line items
 - Auto-fill values year-wise

- Allow manual correction

Dynamic Financial Years

- Replace fixed 5-year view with **dynamic year selector** (1–30 years).
- UI must expand/shrink columns accordingly.

NEW SUBSECTION: STEP 2B – PROJECTIONS MODULE

Insert after Step 2 or create a new **Step 2B**.

4.5.x Projections & Forecasting

1. Projection Input Screen

- Capture projection drivers:
 - Revenue growth %
 - Operating expense %
 - Other configurable drivers

2. Auto-Generated Forecast Statements

- System must auto-generate:
 - Projected Balance Sheet
 - Projected Income Statement
 - Projected Cash Flow (if required)

3. Integration With Scoring

- Projections must influence:
 - Credit Score
 - Derived ratios
 - Final rating

4. Bank-Configurable Projection Rules

- Growth rules & mapping must be fully configurable per bank.

5. Projections & Forecasting

- Projection Input Screen
- Auto-Generated Forecast Statements
- Bank-Configurable Projection Rules
- Historical Qualitative Data Retention

The screenshot displays the 'KivaAI Credit Assessment System' interface. At the top, there's a navigation bar with the KivaAI logo and a 'Credit Risk Assessment' title. Below this, a progress bar shows five steps: 1. Entity Details, 2. Financial Data, 3. Financial Ratios (currently active), 4. Business Factors, and 5. Rating Result. The main content area is titled 'Financial Ratios Assessment' and includes a sub-header 'Template: Contracting & Services Rating'. A 'Scoring' instruction states: 'Rate each parameter 1-5 (Poor to Excellent) based on calculated values'. Three ratio assessment sections are visible: 'Current Ratio', 'Debt to Equity Ratio', and 'Interest Coverage Ratio'. Each section contains a 'Current Ratio' label, a 'Calculated Value' field, an 'Error' field, and a 'Score (1-5)' field with a 'Rate' input. A '100%' progress indicator is shown for each ratio.

Step 3 – Financial Ratios

- Auto-calculated based on formulas defined in Parameters
- Editable if Score Type = Manual
- Shows formula and validation cues

Ratios pulled dynamically based on template parameters.

Types:

- Liquidity
- Leverage
- Profitability
- Efficiency

If formula exists:

- Auto calculate
- Display formula beneath field

Manual override allowed only if Parameter Score Type = Manual.

Client Additions –

Dynamic Ratio Calculation

- Ratios must auto-compute based on:
 - Imported data
 - Edited data
 - Projected financials

Real-Time Updates

- Editing any financial input must instantly update ratios.

Step 4 – Business & Qualitative Factors

- Parameters from template (Qualitative / Score-based)
- Dropdowns, score fields
- Section-wise scoring summaries

Derived from parameters with Data Type = Qualitative/Score.

Dropdown selections map to configured scoring options.

Weighted section score auto-populated from parameter module configuration.

Analyst can input comments or justifications.

Evaluation Criteria Range 1 – 5, for qualitative parameter criteria-wise evaluation and option selection.

Client Additions –

Grouped Questions Support

- Questions must be organized in configurable groups.

Configurable Question Logic

- Admin must configure:
 - Questions
 - Groups
 - Scoring rules
 - Weightages

Miscellaneous Question Subsection

- Add a separate **Miscellaneous Questions** area before final score.

Historical Qualitative Data Retention

- System must retain qualitative answers across assessments and show history.

The screenshot displays the 'Credit Risk Assessment' interface of the KiyaAI Credit Assessment System. The top navigation bar includes the KiyaAI logo and the system name. Below the header, a progress bar shows five steps: 1. Entity Details, 2. Financial Data, 3. Financial Ratios, 4. Business Factors, and 5. Rating Result. The 'Rating Result' step is currently active.

The main content area displays the 'Final Credit Rating' with three key metrics:

- Rating Grade:** D
- Weighted Score:** 0.00
- Default Probability:** 50.00%

Below these metrics is a 'Calculation Breakdown' section. It includes a formula: $\text{Weighted Score} = \sum (\text{Parameter Score} / 5 \times 100 \times \text{Parameter Weight} / 100)$. The 'Final Total Weighted Score' is shown as 0.00, with a note: 'Sum of all weighted scores = 0.00'.

A 'Score Summary' table is also present:

Section	Parameter	Calculated Value	Score	Weight	Weighted Score
TOTAL				0%	0.00

Below the table is an 'Analyst Comments' section with a text input area. At the bottom, there are navigation buttons: 'Previous', 'STATUS: DRAFT', 'Save as Draft', 'Submit for Credit Review', and 'Print Scorecard'.

Rating Result (Step 5)

- System computes:
 - Final Weighted Score

- Rating Grade (e.g., A+, BBB, C)
- Probability of Default (PD)
- Displays:
 - Color-coded rating card
 - Section-wise contribution
 - Full scoring matrix
- Actions:
 - Save
 - Submit for Approval (if workflow enabled)
 - Generate Report (PDF)

Workflow Handling (Optional Maker-Checker) – Will be defined as per Banks requirements

- Maker creates assessment → submits.
- Checker reviews → approves/rejects.
- Audit trail stored for every action.

CIF Tagging

- If tagged later, user can add CIF in edit mode from Dashboard or Assessment List.
- CIF becomes part of customer profile.

Client Additions –

Historical Comparison

- User must be able to:
 - Open previous rating results in popup
 - Compare current vs previous ratings side-by-side

NEW SUBSECTION: MANUAL ADJUSTMENT LAYER

Insert after Rating Result or before Final Submission.

Manual Adjustment Layer

- Only accessible to authorized users (Senior Credit Officer, etc.).
- User can override computed score.
- System must capture:
 - Adjusted value
 - Reason for adjustment

- User
- Timestamp
- All overrides must be included in Audit Trail.

NEW SUBSECTION: DECISION SNAPSHOT

Insert after Manual Adjustment.

Decision Snapshot

At final submission, store a complete snapshot including:

- All financial inputs
- All ratios
- All projections
- All qualitative scores
- Scoring parameters used
- Adjustment logs

Used for:

- Audit
- Regulatory review
- Post-loan monitoring

HIGH-LEVEL INTER-DEPARTMENT WORKFLOW

A. CBD / IBD WORKFLOW (Business Unit)

This unit performs the **pre-risk business checks**, consisting of:

1. Business Maker (X123)

- Creates the case
- Enters all customer/entity details
- Uploads documents
- Saves drafts or submits to checker

Transition → Submits the case to **Business Checker**

2. Business Checker (X123)

- Reviews data entered by Maker

- Can **approve** or **return for corrections**
- Cannot modify actual data (remarks only)

Two possible paths:

- **Return** → Goes back to Business Maker for corrections
- **Submit Forward** → Sends case to **Risk Verifier (RMD)**

This transition to RMD triggers a **system notification**.

B. RISK MANAGEMENT DEPARTMENT (RMD) WORKFLOW

3. Risk Maker (X123)

- Reviews the case sent from CBD/IBD
- Fills Risk Parameters & qualitative risk details
- Uploads risk documents
- Can save draft
- Submits to Risk Checker

Transition → Risk Checker

4. Risk Checker (X123)

- Validates the risk assessment performed by the Risk Maker
- Adds remarks
- Approves or returns for risk-level rework

If approved:

Transition → Final Approver

5. Final Approver (X123)

- Performs final credit-level approval
- Confirms end-to-end completion from Business & Risk
- Once approved, the entire assessment becomes **FROZEN**
- No further edits allowed

End State → Assessment Frozen + Final Rating Report Enabled

(MESSAGE FLOWS)

1. Notification (CBD/IBD → RMD)

Triggered when Business Checker forwards the case to Risk Verifier:

- Risk team receives workflow notification
- Assessment enters Risk Draft state

2. Submission Flow (RMD → CBD/IBD for tracking)

Once Risk Approval completes, the submission progresses to Final Approver:

- Final approval notification
- System locks record post-approval

Risk Management Department (RMD) Workflow

3. Risk Maker (X123)

- Can edit data points entered by business users.

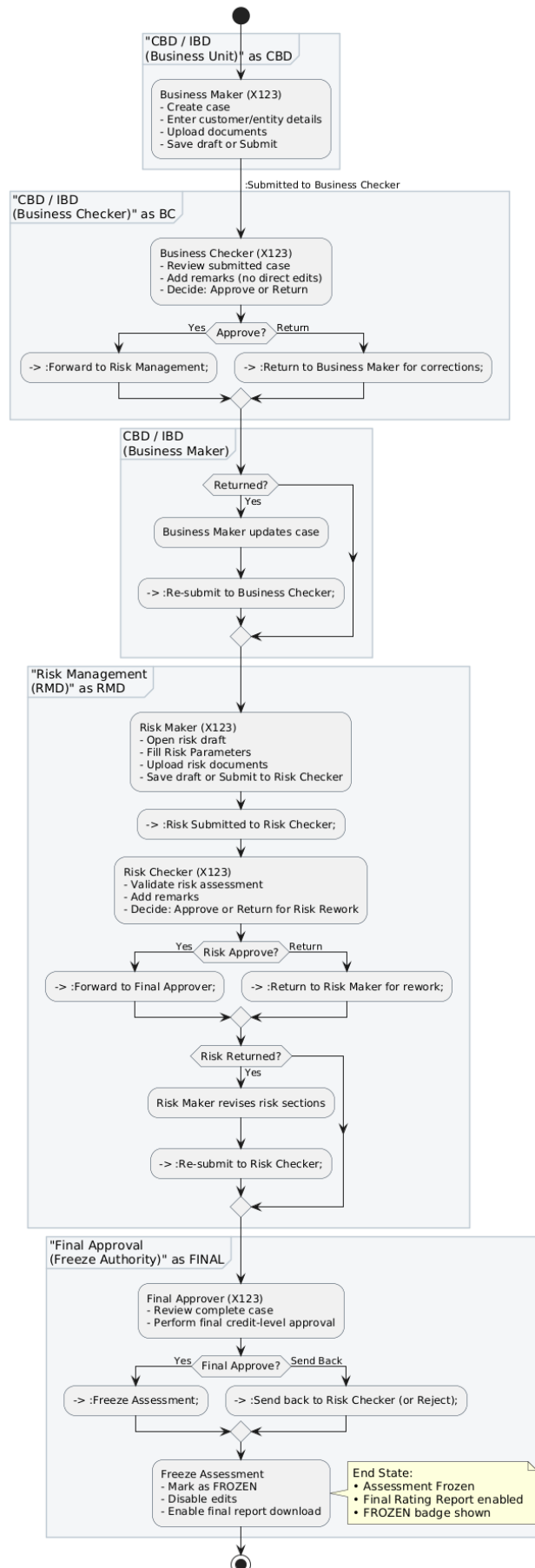
4. Risk Checker (X123)

- Can modify data points.
- Option to return to Risk Maker.

5. Final Approver (X123)

- Can return the application to any Risk user for changes or make changes directly.

KiyaAI Credit Assessment Workflow (Maker → Checker → Risk → Final)



4.6. Reports

Reports
Generate and export credit rating reports

Available Reports + Generate New Report

All Report Types All Status Search by entity name...

REPORT ID	ENTITY NAME	REPORT TYPE	GENERATED DATE	RATING	STATUS	ACTIONS
RPT-2025-001	ABC Manufacturing Ltd.	CREDIT RATING	Nov 10, 2025	AAA	COMPLETED	
RPT-2025-002	XYZ Retail Corporation	FINANCIAL ANALYSIS	Nov 9, 2025	BBB+	COMPLETED	
RPT-2025-003	Tech Innovations Pvt. Ltd.	RISK ASSESSMENT	Nov 8, 2025	AA-	COMPLETED	
RPT-2025-004	Global Logistics Inc.	CREDIT RATING	Nov 7, 2025	A+	COMPLETED	
RPT-2025-005	Green Energy Solutions	PORTFOLIO SUMMARY	Nov 6, 2025	BBB	PENDING	
RPT-2025-006	Prime Construction Ltd.	CREDIT RATING	Nov 5, 2025	A-	DRAFT	

Reports
Generate and export credit rating reports

Available Reports + Generate New Report

All Report Types All Status Search by entity name...

REPORT ID	ENTITY NAME	REPORT TYPE	GENERATED DATE	RATING	STATUS	ACTIONS
RPT-2025-001	ABC Manufacturing Ltd.	CREDIT RATING	Nov 10, 2025	AAA	COMPLETED	
RPT-2025-002	XYZ Retail Corporation	FINANCIAL ANALYSIS	Nov 9, 2025	BBB+	COMPLETED	
RPT-2025-003	Tech Innovations Pvt. Ltd.	RISK ASSESSMENT	Nov 8, 2025	AA-	COMPLETED	
RPT-2025-004	Global Logistics Inc.	CREDIT RATING	Nov 7, 2025	A+	COMPLETED	
RPT-2025-005	Green Energy Solutions	PORTFOLIO SUMMARY	Nov 6, 2025	BBB	PENDING	
RPT-2025-006	Prime Construction Ltd.	CREDIT RATING	Nov 5, 2025	A-	DRAFT	

The Reports module allows users to **generate, download, and review credit assessment reports**. It provides standardized document formats based on templates, offering transparency and auditability of rating results.

The Reports screen includes:

A. Report List Table

Columns:

- Report Name
- Description
- Report Type (PDF/Template-Based)
- Last Generated
- Actions (Generate, Download, View)

B. Report Generation Filters

For applicable reports:

- Date Range
- Status
- Template
- Entity Name
- Financial Year

These filters refine the dataset before generating output.

C. Output Format

Reports generated in:

- PDF
- Excel (if defined)
- System-defined Template Output

Types of Reports

Common reports include:

- **Assessment Summary Report**
- **Parameter Score Report**
- **Final Rating Report**
- **Portfolio Overview Report**
- **Assessment Status Report**

Report Generation Process

- User selects the report type.
- Applies filters based on available input parameters.
- System fetches relevant data set.
- Backend prepares the report using predefined template.
- User downloads or previews the output.

Report Template Management

- System supports configuration of:
 - Report Layout
 - Header/Footer
 - Branding

- Section Arrangement
- Admin module controls available templates (future scope if not yet implemented).

Sample Report –

Portfolio-Level Assessment Report

[illegible]

This report represents a Customer-Level Assessment under the *Contracting and Services* credit scoring model.

Each row corresponds to one customer (CIF) and shows the scores assigned across all qualitative, quantitative, and mandatory adjustment parameters defined in the model.

1. Customer Information

- CIF – Customer Identification File number (unique customer ID)
- Name – Customer name

These identify the customer for whom the assessment is performed.

2. Qualitative Parameters (Company & Business Factors)

The first set of columns captures non-financial, qualitative factors that influence customer risk. Examples:

- Constitution & Establishment – Type of entity (Govt, Public Co., Partnership, etc.)
- Number of Years in Business
- Credit Line Utilization with Other Lenders
- Financial Planning / Internal Controls
- Market Position
- Sector Contracting (Govt./Private Sector)
- Sensitivity to Economic Shocks
- Contractor Grade
- Past Track Record
- Competition & Demand
- Availability of Qualified Resources

Each factor shows a score based on predefined risk weight tables.

3. Quantitative Parameters (Financial Ratios)

The green columns reflect financial ratio scoring, including:

- Affiliate Due to Net Worth
- Current Ratio
- Debt-Equity Ratio
- Operating Revenue / External Liabilities
- Investment Assets to Total Assets
- Net Profit Margin
- Working Capital Turnover
- Change in Debt Ratio

These values are calculated using customer's financial statements and mapped to the scoring ranges you provided earlier.

In this screenshot, financial numbers appear as NA, meaning:

- Financials were not available OR
- Customer belongs to a purely qualitative model OR
- This is sample dummy data

4. Miscellaneous Factors

These additional risk elements include:

- Account Conduct – Evergreen accounts, repayment behavior
- Economic Shock Factors – Sensitivity to market downturns, oil price crash, etc.
- # of Notches – Adjustment for overrides (manual upgrades/downgrades)

These influence the overall score when applicable.

5. Mandatory Adjustments

Mandatory risk modifiers include:

- Age of Financial Information
- Quality of Financial Information
- Historical Account Performance
- Qualification of Audit Report

These ensure that low-quality or outdated financials penalize the final outcome.

6. Final Score

The final column provides the total aggregated score after:

Qualitative score

Quantitative score

Miscellaneous

Mandatory Adjustments

In the example:

- CIF 001 → Final Score = 128
- CIF 002 → Final Score = 233

Client Additions -

Include Decision Snapshot in reports.

Add **Historical Rating Comparison Report**.

Include projection-based statements in Assessment Report.

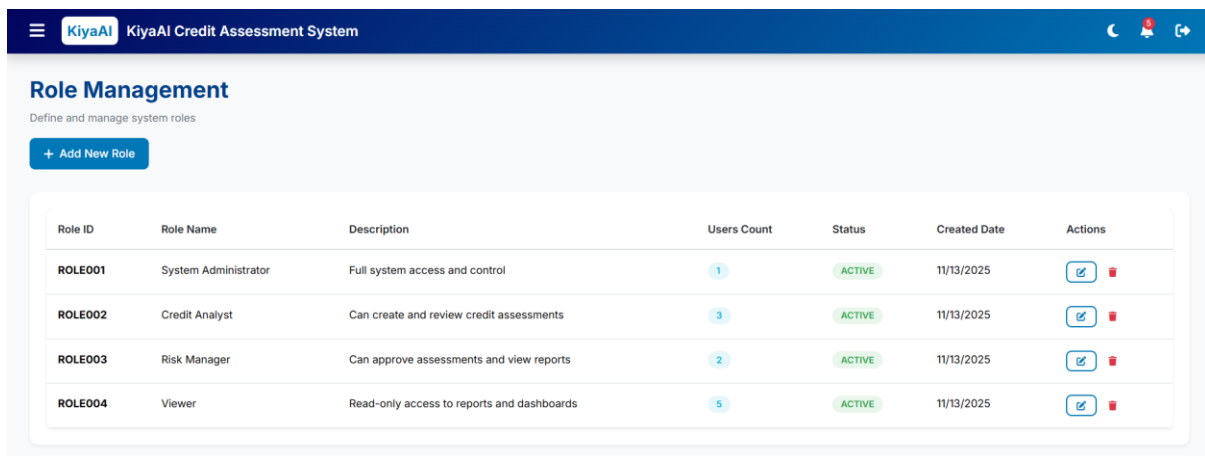
The system shall generate two reports:

1. **Customer-Level Assessment Report** – showing individual customer scores across all applicable models.
2. **Portfolio-Level Assessment Report** – showing aggregated scores and summaries across the entire customer portfolio.

Sample reports will be provided for all customers across all models with their respective scores.

4.7. Administration

4.7.1. Role Management



Role ID	Role Name	Description	Users Count	Status	Created Date	Actions
ROLE001	System Administrator	Full system access and control	1	ACTIVE	11/13/2025	Edit Deactivate Delete
ROLE002	Credit Analyst	Can create and review credit assessments	3	ACTIVE	11/13/2025	Edit Deactivate Delete
ROLE003	Risk Manager	Can approve assessments and view reports	2	ACTIVE	11/13/2025	Edit Deactivate Delete
ROLE004	Viewer	Read-only access to reports and dashboards	5	ACTIVE	11/13/2025	Edit Deactivate Delete

The Roles module enables administrators to create and manage **user roles** within the application. Each role defines a set of permissions that control access to modules, screens, and actions across the system.

It forms the foundation of RBAC (Role-Based Access Control).

The Role Management screen displays:

A. Roles List Table

Includes:

- Role ID
- Role Name
- Role Type (System / Custom)
- Description
- Status (Active / Inactive)
- Actions (Edit, Activate/Deactivate)

B. Add/Edit Role Modal

Fields:

- Role Name
- Description
- Role Status
- Permissions Assignment (linked with RBAC module)
- Save / Cancel buttons

FUNCTIONAL DETAILS

Create Role

- Admin enters:
 - Role Name (e.g., Analyst, Reviewer, Admin)
 - Description
 - Status
- Saves role to system.
- Default roles may be predefined (e.g., System Admin).

Edit Role

- Admin can update:
 - Name
 - Description
 - Status
- Permissions edited via the RBAC module.

Role Status Management

- **Active** roles available for assignment.
- **Inactive** roles cannot be assigned.
- Audit logs maintained for status changes.

Permissions Integration

- Role definition alone does not grant access.
- Permissions must be linked in **RBAC Module**.
- Each role maps to modules, screens, or actions.

Validation Rules

- Role name must be unique.
- Cannot deactivate role if users are actively assigned (unless allowed by business).

4.7.2. User Management

User Management
Manage system users and their roles

[+ Add New User](#)

All Users All Roles

User ID	Full Name	Email	Role	Department	Status	Last Login	Actions
1	undefined undefined	admin@kiyaai.com	UNDEFINED	-	ACTIVE	11/12/2025, 10:56:52 AM	Edit Deactivate
2	undefined undefined	john.maker@kiyaai.com	UNDEFINED	-	ACTIVE	11/12/2025, 8:30:00 AM	Edit Deactivate
3	undefined undefined	sarah.checker@kiyaai.com	UNDEFINED	-	ACTIVE	11/12/2025, 9:15:00 AM	Edit Deactivate
4	undefined undefined	mike.analyst@kiyaai.com	UNDEFINED	-	ACTIVE	11/11/2025, 4:45:00 PM	Edit Deactivate

The User Management module enables administrators to create, update, and maintain user accounts for the KiyaAI Credit Assessment System.

It defines user identity, assigned roles, access levels, and operational status.

This module ensures secure and controlled access of users across the system.

The User Management screen includes:

A. User List Table

Columns:

- User ID
- Name
- Email
- Role
- Status (Active / Inactive / Locked)
- Last Login
- Actions (Edit, Activate/Deactivate, Reset Password)

B. Add/Edit User Form/Modal

Fields:

- Full Name
- Email Address
- Phone Number (optional)
- Role Assignment (dropdown)
- User Type (ADS / Manual Login)
- Status

- Optional: Department / Location
- Save / Cancel

C. Optional Actions

- **Reset Password** (Manual login users)
- **Unlock User** (after failed attempts logout)
- **Deactivate Account**

FUNCTIONAL DETAILS

Create User

- Admin enters user details.
- Selects **Role** from Role Management module.
- Selects login type:
 - **ADS User** → Authenticated through Active Directory
 - **Manual User** → Password generated and sent via email
- Saves user into system.

Edit User

- Admin updates user details:
 - Name
 - Email
 - Role
 - Status
- Critical fields like email may be restricted depending on business rules.

User Status Management

Statuses include:

- **Active** → User can log in
- **Inactive** → User blocked from login
- **Locked** → System-locked due to failed attempts

Admin can:

- Unlock user
- Activate / Deactivate account

Password & Authentication Handling

For Manual Login Users:

- Admin can reset password
- User receives system-generated credentials
- User must change password on first login (optional requirement)

For ADS Users:

- Password managed in Active Directory
- No reset option inside system

Audit Trail Integration

System logs:

- User creation
- Updates
- Role changes
- Status changes
- Login attempts
- Failed login events

4.7.3. RBAC (Role-Based Access Control)

Role Based Access Control
Configure module permissions for each role

Configure Access Permissions

Select Role *
System Administrator

Workflow Type
None

Module Category
All Modules

Module Name	View	Add	Edit	Delete	Export	Approve
Dashboard	☒	☒	☒	☒	☒	☒
Credit Assessment	☒	☒	☒	☒	☒	☒
Rating Templates	☒	☒	☒	☒	☒	☒
Parameters Management	☒	☒	☒	☒	☒	☒
Reports	☒	☒	☒	☒	☒	☒

The RBAC (Role-Based Access Control) module defines **permissions and access rights** for each role in the system.

It determines what every role can **view, edit, create, approve, or delete** across modules.

RBAC ensures users only access features relevant to their responsibilities.

The RBAC screen includes:

A. Role Selection Panel

- Dropdown listing all active roles
- Selecting a role loads its current permissions

B. Permissions Matrix

Displayed in a table/grouped list:

- Modules (Dashboard, Assessment, Templates, Parameters, Reports, Admin, etc.)
- Submodules and screens
- Action-level controls, such as:
 - View
 - Create
 - Edit
 - Approve
 - Delete
 - Download Reports
 - Manage Workflow

Each permission uses toggle switches or checkboxes.

C. Save Permissions

- Button to persist changes for the selected role.

FUNCTIONAL DETAILS

Permission Assignment

For each role, admin can enable/disable access to:

- Entire modules
- Specific screens
- Field-level or action-level permissions (if configured)

Example:

- Analyst → View + Edit Assessments
- Reviewer → View + Approve Assessments
- Admin → Full Access

Permission Types

Common permission categories:

- **Module Access** (visibility in sidebar)
- **Screen Access**
- **Create/Edit/Delete**
- **Approval Rights**
- **Report Generation**
- **Admin-Level Controls**

Enforcement Mechanism

- Permissions are stored against each role.
- At login:
 - System loads user's role and permission set
 - Sidebar and screens dynamically adjust
- Actions blocked by API if unauthorized.

Integration with User Management

- Users inherit permissions from assigned roles.
- Role change automatically updates user access.

Audit Tracking

System logs:

- Permission modifications
- Modified by
- Timestamp

4.8. Masters

4.8.1. Lookup Management

Lookup Management + New Lookup

Pending (12) Authorized (76) Rejected (7)

Q Type to search and filter data across all columns in the table

Lookup ID	Lookup Type	Lookup Type Code	Description	Created By	Application Date	Last Assessed By	Action
63	Regions	System	Regions Description	Feb 10, 2025, 1:16:36 PM			
64	Zone	System	Zone Description	Feb 10, 2025, 1:17:05 PM	Feb 10, 2025, 1:16:36 PM		
65	Financial Months	System	Financial Months Description	Feb 10, 2025, 1:17:22 PM	Chetan Sharma	Aug 18, 2025, 1:08:4	
66	Branch Working Week Days	System	Branch Working Week Days Description	Feb 10, 2025, 1:17:36 PM			
67	Time Zone	System	Time Zone Description	Feb 10, 2025, 1:17:49 PM			
68	Blood Groups	System	Blood Groups	Feb 10, 2025, 1:18:02 PM			
69	Workflow	System	Workflow	Feb 10, 2025, 1:18:16 PM	Chetan Sharma	Sep 1, 2025, 11:55:3	
71	Time Format	System	Time Format Description	Feb 10, 2025, 1:18:51 PM			

Add New Lookup

Lookup Type * Lookup Type Code * Lookup Type Description *

Lookup Details

Lookup Type Code *	Display Value *	Action
Lookup code	Lookup Display Value	

Add

Cancel Submit

The Lookup Management module allows administrators to maintain all **static and reusable dropdown values** used across the system.

These lookups ensure consistency in data entry and support configurable options across multiple modules (Assessment, Templates, Parameters, Reports, etc.).

The Lookup Management screen includes:

A. Lookup Category List

Examples:

- Industry Category
- Entity Type
- Business Risk Options
- Qualitative Score Options
- Assessment Status
- Document Types
- Rating Grades
- Financial Ratio Types

B. Lookup Value Table

Columns:

- Lookup ID
- Lookup Name
- Display Value
- Status (Active/Inactive)
- Actions (Edit/Delete)

C. Add/Edit Lookup Modal

Fields:

- Category
- Lookup Name
- Display Value
- Description (optional)
- Status
- Save/Cancel buttons

FUNCTIONAL DETAILS

Create Lookup

- Admin selects a category or creates a new category.
- Adds lookup name and display value.
- Saves to system for immediate availability.

Edit Lookup

- Modify existing lookup:

- Name
 - Display value
 - Status
- Deactivated lookups stop appearing in dropdowns.

Lookup Categories

Each category groups related values. Examples:

- **Industry Sector** for Entity Details
- **Template Categories** for Rating Templates
- **Qualitative Options** for Parameters
- **Assessment Status Types** for workflow

System Usage

Lookups populate:

- Dropdowns
- Filter lists
- Auto-complete inputs
- Validation reference data

Changes reflect across modules instantly.

Validation Rules

- Lookup name must be unique within its category.
- Category cannot be deleted if values exist.
- Inactive values cannot be assigned in new records.

4.9. System Configuration

The System Configuration module allows administrators to manage global system settings that impact communication, module visibility, and operational behavior. It includes configuration of **Email Server**, **Email Templates**, and **Module Management**.

Module Management

+ New Module

Pending (3) **Authorized (11)** Rejected (0)

Q Type to search and filter data across all columns in the table

Module ID	Module Name	Status	Created By	Application Date	Last Assessed By	Last Assessed Date	Action
3	Masters	Active	Jan 9, 2025, 11:34:59 AM	Chetan Sharma	Nov 4, 2025, 12:28:23 PM		View Edit Delete
4	Administration	Active	Jan 9, 2025, 11:35:15 AM	Chetan Sharma	Oct 27, 2025, 12:29:17 PM		View Edit Delete
18	System Configurations	Active	Jun 9, 2025, 11:32:43 AM	Nov 4, 2025, 12:28:23 PM			View Edit Delete
36	Document Management	Active	Aug 11, 2025, 5:13:55 PM	Chetan Sharma	Nov 7, 2025, 12:35:36 PM		View Edit Delete
37	Document Templates	Active	Chetan Sharma	Aug 12, 2025, 3:45:13 PM	Chetan Sharma	Nov 4, 2025, 12:28:23 PM	View Edit Delete
38	Repository Management	Active	Chetan Sharma	Aug 12, 2025, 3:50:23 PM	Chetan Sharma	Nov 5, 2025, 4:12:28 PM	View Edit Delete
39	Versioning and History	Active	Chetan Sharma	Aug 12, 2025, 3:54:14 PM	Chetan Sharma	Nov 4, 2025, 12:28:23 PM	View Edit Delete
43	Tags Management	Active	Chetan Sharma	Aug 20, 2025, 11:30:26 AM	Chetan Sharma	Nov 5, 2025, 4:12:28 PM	View Edit Delete

Add New Module

Module Details

Module Name * Description * SVG ICON *

Module Name Module Description SVG Path Icon

Sub Module Details

Add Sub Module

Sub Module Name	Sub Module URL	Merge Tags	Action
Sub Module Name	Url eg. /customer-management	Select Tags	Delete

☐ Would you like to add components to sub module?

Cancel

Submit

Email Configuration

SMTP ID * SMTP Host * SMTP Port *

SMTP ID SMTP Host SMTP Port

SMTP Username * SMTP Password * Protocol *

SMTP Username SMTP Password Protocol

SMTP Properties

SMTP Auth * SMTP Timeout * SMTP Connection Timeout *

SMTP Auth SMTP Timeout SMTP Connection Timeout

SMTP StartTLS Properties

SMTP StartTLS Enabled * SMTP StartTLS Required *

SMTP StartTLS Enabled SMTP StartTLS Required

Update

Submit

View Email Template Pending Close

Template Code

TEMP_1234

Template Name

DMS Document Upload

Email Subject

DMS Document Uploaded Successfully

Template Module

3 - Masters

Template Sub Module

4 - Country Management

Direction

Forward

Trigger when,

Country Status : Updated, Authorised

Template Tags

Sr.	Code	Tags
1	countryCode	{{COUNTRY_CODE}}
2	countryName	{{COUNTRY_NAME}}
3	countryAbb	{{COUNTRY_ABBREVIATION}}
4	isdCode	{{COUNTRY_ISD_CODE}}
5	zone	{{COUNTRY_ZONE}}

Template Body *

?? Document Uploaded Successfully!

Hello {{COUNTRY_CODE}},

Your document has been successfully uploaded to the DMS platform. You can now access, review, or manage it directly from your dashboard.

?? Document Name:

Financial Report_Q3_2025.pdf

Columns

Button

Divider

Heading

Text

Image

Social

Menu

HTML

Content

Blocks

Body

by Unlayer Editor

Cancel

Submit

A. Email Configuration

Fields:

- SMTP Server
- Port
- Sender Email (From Address)
- Username
- Password / App Key
- SSL/TLS Enable Toggle

Email Configuration

- Admin enters SMTP details required for system notifications.
- “Test Email” validates configuration.
- Used for:
 - User creation emails
 - Password reset
 - Assessment alerts

- Report delivery (if applicable)

Validation:

- Mandatory fields must be valid before save.

B. Email Templates

Columns:

- Template Name
- Subject
- Type (Password Reset, Notification, Assessment Report, Alerts)
- Actions (Edit/Preview)

Add/Edit Template Modal:

- Template Name
- Subject Line
- Body (Rich Text Editor)
- Dynamic Variables (e.g., {{UserName}}, {{AssessmentId}})
- Admin manages system-generated email formats.
- Templates support placeholders for dynamic data.
- Typical templates:
 - User Onboarding
 - Password Reset
 - Assessment Completed
 - Approval Notifications

Editing:

- Admin updates subject and body via rich text editor.
- Preview option before save.

C. Module Management

- Toggle-based list of modules:
 - Dashboard
 - Credit Assessment
 - Templates
 - Parameters

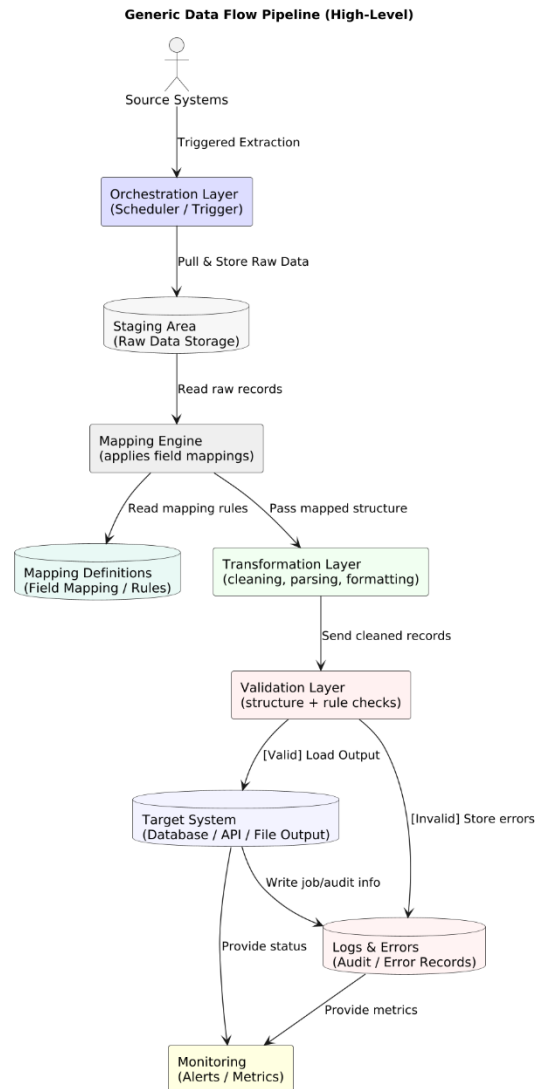
- Reports
 - Audit
 - Admin Modules
- Admin can **enable/disable** modules for all users or selected roles (if extended).

Controls which modules appear in the sidebar.

Disabling a module hides it from all users (unless role-based override is configured).

Helps manage phased rollouts or features under development.

5. Data Ingestion



High-Level Stepwise Summary

1. Extract

Data is pulled from source systems (DB, API, files) based on a trigger or schedule.

2. Stage

Raw data is stored temporarily without changes, to keep an untouched copy.

3. Map

A mapping engine reads mapping definitions and aligns source fields to target fields.

4. Transform

Mapped data is cleaned, formatted, and converted into the required structure.

5. Validate

Data is checked for correctness, mandatory fields, formats, and business rules.

6. Load

Valid data is loaded into the target system (database, file, API, etc.).

7. Log & Error Handling

Invalid data and process errors are recorded for debugging or reprocessing.

8. Monitor

System monitors job status, performance, and alerts on failures or anomalies.

Financial Data Ingestion & Auto-Population

The Credit Assessment System requires complete and accurate financial information to calculate ratios, scores, and the final borrower rating.

Since the Core Banking System (CBS) does not store detailed financial statements, the system supports **three user-friendly methods** to bring audited financial data into the rating workflow:

1. **Excel Upload (Primary & Preferred Method)**
2. **OCR Upload (PDF/Image Recognition)**
3. **Manual Entry (Fallback Method & NEW Customers)**

Regardless of the method used, the system automatically fills the financial forms, computes all ratios using predefined formulas, and calculates the final rating in real-time.

A. What Data Comes From CBS

When the user enters a CIF, the system automatically fetches the following CBS information:

- Customer Name
- Entity Type
- Industry / Sector
- CINET number
- Customer Status (Active, Closed, etc.)
- Loan Accounts (Account ID, Type, Sanctioned Limit, Outstanding, Overdues)
- Collateral Details (if available)

CBS does NOT provide: Revenue, Assets, Liabilities, EBITDA, Inventory, Profit, Borrowings break up, etc.

These must come from Excel/OCR or manual entry.

CBS data is used only to identify the customer and show current exposure.

B. Staging Area Approach

Because CBS is owned by another vendor, all CBS data first comes into a **secure staging area** before being used.

Functionally, the staging area works like a **mailbox**:

1. CBS places customer and loan data in the staging folder.
2. The credit system picks the files from staging.
3. System fills customer details automatically in Step-1 of the Assessment.

The staging area ensures:

- Clean separation between CBS and our system
- Full audit trails
- Simple, secure data exchange
- No need for CBS vendor changes

C. Excel Upload – Primary Method for Financial Statements

This is the main method used by the bank because Excel provides the cleanest audited values.

Step-by-Step Functional Flow

1. User uploads the bank's standard Excel template.
2. System checks:
 - Is format correct?
 - Are all required fields present?
 - Are the numbers valid?
3. System reads every financial value from the Excel:
 - Revenue
 - COGS
 - EBITDA
 - Net Profit
 - Current Assets / Liabilities
 - Inventory
 - Receivables / Payables
 - Short-term & Long-term Borrowings
 - Total Assets / Liabilities
 - Equity
 - Interest Expense
4. System automatically fills all these values into:

- Balance Sheet Section
 - Profit & Loss Section
 - Cash Flow Section (if available)
5. User does NOT need to type anything manually.
 6. All fields remain editable so the analyst can correct or adjust values.

D. OCR Upload – If Excel Is Not Available

If the user uploads a **PDF or scanned image** of financial statements:

1. The OCR engine reads values directly from the PDF.
2. Extracted values appear on screen for review.
3. User can correct any mistakes.
4. Once confirmed, the system treats it the same as Excel upload.

OCR ensures the bank can still proceed with assessments even if the customer only gives a PDF.

E. Manual Entry – Fallback Method

If no Excel or PDF is available:

1. User manually enters financial values.
2. System immediately calculates ratios & scoring.
3. All inputs remain editable.

This ensures assessments can continue without delay.

F. Auto-Population of Assessment Form

Once financial data is received through Excel, OCR, or manual entry, the system:

1. Automatically fills:

- Balance Sheet
- P&L Statement
- Cash Flow Statement
- Financial Projections (if enabled)
- All parameter values used in the rating template

2. No retyping required

Every financial input field is pre-filled.

3. Every field remains editable

If the analyst changes any number:

- All ratios update instantly
- All scores recalculate automatically
- Final rating adjusts in real time

G. Ratio Calculation – Explanation

The system uses standard banking formulas.

Examples:

Current Ratio

= Current Assets ÷ Current Liabilities

Quick Ratio

= (Current Assets – Inventory) ÷ Current Liabilities

Debt/Equity Ratio

= Total Debt ÷ Shareholder Equity

Interest Coverage Ratio

= EBITDA ÷ Interest Expense

Working Capital Cycle

= Receivable Days + Inventory Days – Payable Days

All these values are calculated **automatically** by the system.

No user calculation is required.

H. Scoring & Final Rating – Simple Explanation

1. Each ratio has a scoring rule based on bank thresholds.

Example:

- Current Ratio > 2.0 → Score: 4
- Current Ratio < 1.0 → Score: 1

2. Each parameter has weightage defined in the Rating Template.

Example:

- Current Ratio Weight = 5%
- Profitability Weight = 10%

3. System applies the weightage to each score.
4. Total weighted score is calculated.
5. Final Score maps to rating grade (A+, A, BBB, BB, etc.)

This calculation is **fully automated**.

I. Editable Inputs & Audit Trail

Editable Inputs

- All imported values (Excel/OCR)
- All manually entered values
- All projected values

Non-Editable Inputs

- System-calculated ratios
- System-calculated weighted scores
- Final rating grade (unless adjusted in “Manual Adjustment Layer”)

J. End-to-End Example (Functional)

1. User enters CIF → system fetches customer info from CBS.
2. User uploads Excel for FY 2023.
3. System reads all 50+ financial fields.
4. System fills the entire financial form automatically.
5. System calculates:
 - 15+ ratios
 - Section scores
 - Weighted contributions
 - Final credit rating
6. User edits one value (e.g., Inventory).
7. System instantly recalculates ratios, scores, and rating.
8. User submits assessment for approval.
9. System stores a complete snapshot for audit.

K. Why This Approach Works for the Bank (Business Benefits)

- No dependency on CBS for financials
- No manual calculations needed
- Faster assessment turnaround
- Cleaner, standardized financial data
- All ratios & ratings fully automated
- Supports Maker–Checker–Approver workflow

- Full audit trail for compliance
- Supports both new and existing customers
- Supports projections & forecasting
- Works even when only PDFs exist (OCR fallback)

MASTER FINANCIAL RATIO TABLE

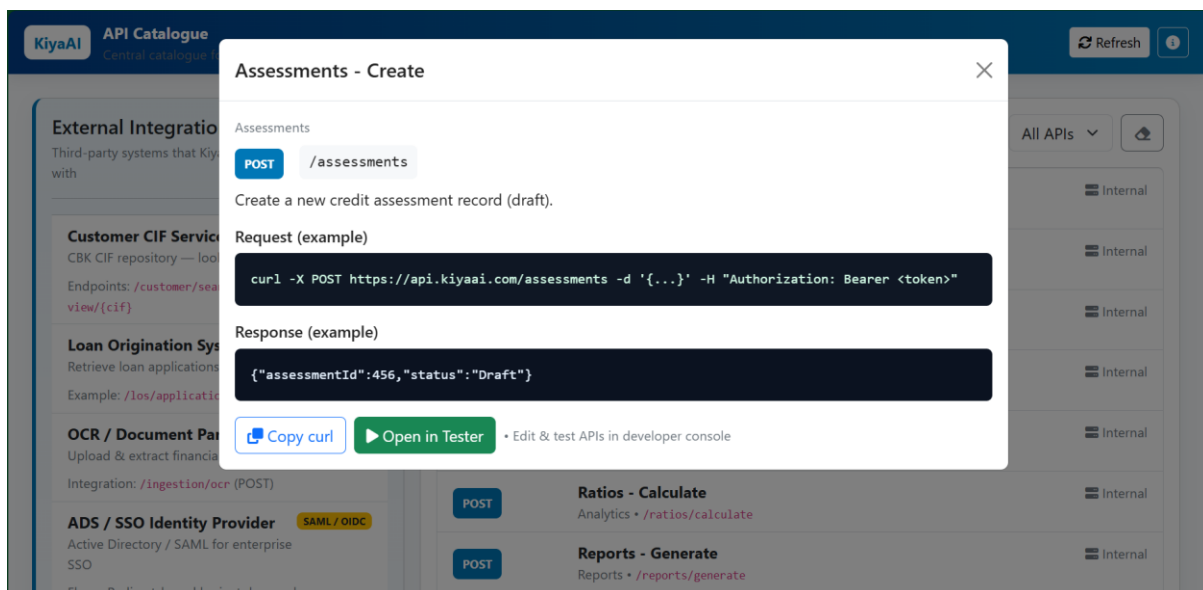
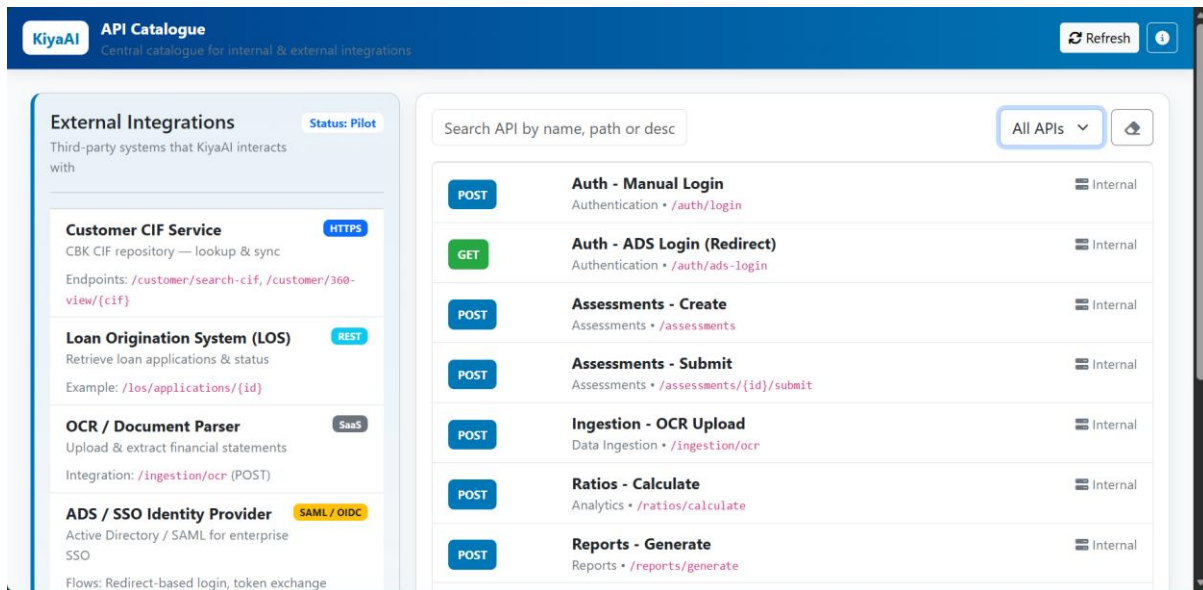
Ratio Name	Formula	Required Fields	Category	Bank Requires Yes/No
Current Ratio	Current Assets / Current Liabilities	Current Assets, Current Liabilities	Liquidity	
Quick Ratio	(CA – Inventory) / CL	Current Assets, Inventory, Current Liabilities	Liquidity	
Cash Ratio	Cash & Equivalents / CL	Cash & Cash Equivalents, Current Liabilities	Liquidity	
Working Capital	CA – CL	Current Assets, Current Liabilities	Liquidity	
Working Capital Ratio	WC / Total Assets	Current Assets, Current Liabilities, Total Assets	Liquidity	
Debt to Equity	Total Debt / Equity	Short-Term Borrowings, Long-Term Borrowings, Equity	Leverage	
Total Liabilities to Total Assets (TLTA)	Total Liabilities / Total Assets	Total Liabilities, Total Assets	Leverage	
Debt/EBITDA	Total Debt / EBITDA	Short-Term Borrowings, Long-Term Borrowings, EBITDA	Leverage	
Debt/Capital	Total Debt / (Debt + Equity)	Total Debt, Equity	Leverage	
Interest-Bearing Debt Ratio	Interest-Bearing Debt / Total Assets	Interest-Bearing Debt, Total Assets	Leverage	
Interest Coverage Ratio (ICR)	EBITDA / Interest Expense	EBITDA, Interest Expense	Leverage	

Fixed Charge Coverage Ratio (FCCR)	$(\text{EBITDA} - \text{CAPEX}) / (\text{Interest} + \text{Principal Due})$	EBITDA, CAPEX, Interest Expense, Principal Due	Leverage	
LT Debt to Capitalization	$\text{LT Borrowings} / (\text{LT Borrowings} + \text{Equity})$	LT Borrowings, Equity	Leverage	
Gross Profit Margin	$(\text{Revenue} - \text{COGS}) / \text{Revenue}$	Revenue, COGS	Profitability	
EBITDA Margin	$\text{EBITDA} / \text{Revenue}$	EBITDA, Revenue	Profitability	
Operating Margin (EBIT Margin)	$\text{EBIT} / \text{Revenue}$	EBIT, Revenue	Profitability	
Net Profit Margin	$\text{NPAT} / \text{Revenue}$	NPAT, Revenue	Profitability	
ROA	$\text{NPAT} / \text{Total Assets}$	NPAT, Total Assets	Profitability	
ROE	$\text{NPAT} / \text{Equity}$	NPAT, Equity	Profitability	
ROCE	$\text{EBIT} / (\text{Total Assets} - \text{CL})$	EBIT, Total Assets, Current Liabilities	Profitability	
ROIC	$\text{NOPAT} / \text{Invested Capital}$	NOPAT, Invested Capital	Profitability	
Asset Turnover	$\text{Revenue} / \text{Total Assets}$	Revenue, Total Assets	Profitability	
Inventory Turnover	$\text{COGS} / \text{Inventory}$	COGS, Inventory	Efficiency	
Receivables Turnover	$\text{Revenue} / \text{AR}$	Revenue, Accounts Receivable	Efficiency	
Payables Turnover	COGS / AP	COGS, Accounts Payable	Efficiency	
Receivable Days	$(\text{AR} / \text{Revenue}) \times 365$	Accounts Receivable, Revenue	Efficiency	
Inventory Days	$(\text{Inventory} / \text{COGS}) \times 365$	Inventory, COGS	Efficiency	
Payable Days	$(\text{AP} / \text{COGS}) \times 365$	Accounts Payable, COGS	Efficiency	
Working Capital Cycle	$\text{Rec Days} + \text{Inv Days} - \text{Pay Days}$	AR, Inventory, AP, Revenue, COGS	Efficiency	
Fixed Asset Turnover	$\text{Revenue} / \text{Net Fixed Assets}$	Revenue, Net Fixed Assets	Efficiency	
Capital Employed Turnover	$\text{Revenue} / \text{Capital Employed}$	Revenue, Capital Employed	Efficiency	
CFO to Debt	$\text{CFO} / \text{Total Debt}$	CFO, Total Debt	Cash Flow	

Cash Conversion Ratio	$\text{CFO} / \text{EBITDA}$	CFO, EBITDA	Cash Flow	
Free Cash Flow	$\text{CFO} - \text{CAPEX}$	CFO, CAPEX	Cash Flow	
FCF to Debt Service	$\text{FCF} / (\text{Interest} + \text{Principal})$	FCF, Interest Expense, Principal Due	Cash Flow	
DSCR	$(\text{EBITDA} - \text{Taxes} - \text{WC Changes}) / (\text{Interest} + \text{Principal})$	EBITDA, Taxes, WC Changes, Interest, Principal Due	Cash Flow	
Operating Cash Flow Ratio	$\text{CFO} / \text{Current Liabilities}$	CFO, Current Liabilities	Cash Flow	
Equity Ratio	$\text{Equity} / \text{Total Assets}$	Equity, Total Assets	Capital Structure	
Debt Ratio	$\text{Total Debt} / \text{Total Assets}$	Total Debt, Total Assets	Capital Structure	
Financial Leverage Index	ROE / ROA	ROE, ROA	Capital Structure	
Contract Profitability Ratio	$\text{Contract Profit} / \text{Contract Revenue}$	Contract Profit, Contract Revenue	Contracting	
Cost to Complete (%)	$\text{Remaining Cost} / \text{Total Estimated Cost}$	Remaining Cost, Total Estimated Cost	Contracting	
Billing Efficiency	$\text{Billed Amount} / \text{Work Completed}$	Billed Amount, Work Completed	Contracting	
Order Book to Revenue	$\text{Order Book} / \text{Revenue}$	Order Book, Revenue	Contracting	
Advance Utilization	$\text{Unutilized Advances} / \text{Total Advances}$	Unutilized Advances, Total Advances	Contracting	
Lender's Case DSCR	$\text{CFA} / \text{Debt Service}$	Cash Flow Available, Debt Service	Project Finance	
Interest Reserve Coverage	$\text{Interest Reserve} / \text{Interest Requirement}$	Interest Reserve, Interest Requirement	Project Finance	
Bank Turnover Ratio	$\text{Bank Turnover} / \text{Sales}$	Bank Turnover, Revenue	SME/MSME	
GST Turnover Ratio	$\text{GST Turnover} / \text{Audited Turnover}$	GST Turnover, Audited Turnover	SME/MSME	
Promoter Contribution Ratio	$\text{Promoter Contribution} / \text{Project Cost}$	Promoter Contribution, Project Cost	SME/MSME	

Net Working Capital Gap	WC Requirement – NWC	WC Requirement, Net Working Capital	SME/MSME	
Revenue Growth %	(CY Revenue – PY Revenue) / PY Revenue	Current Year Revenue, Previous Year Revenue	Growth	
Profit Growth %	(CY NPAT – PY NPAT) / PY NPAT	NPAT CY, NPAT PY	Growth	
EBITDA Growth %	(CY EBITDA – PY EBITDA) / PY EBITDA	EBITDA CY, EBITDA PY	Growth	
Asset Growth %	(Assets CY – Assets PY) / Assets PY	Total Assets CY, Total Assets PY	Growth	
Security Coverage Ratio	Realizable Security Value / Outstanding Exposure	Realizable Security Value, Outstanding Exposure	Risk	
Loan-to-Value (LTV)	Loan Amount / Asset Value	Loan Amount, Asset Value	Risk	
DP Utilization	Outstanding / Drawing Power	Outstanding, Drawing Power	Risk	
Limit Utilization	Avg Utilization / Sanction Limit	Average Utilization, Sanction Limit	Risk	

6. API Catalogue



The API Catalog defines all backend services that the KiyaAI Credit Assessment System exposes or consumes.

These APIs are used by the frontend, admin modules, workflow engine, data ingestion components, and external systems where integration is required.

APIs ensure that all modules interact consistently and securely through REST-based communication.

The API Catalogue dashboard consists of the following sections:

A. External APIs Panel

This section lists all **third-party or client-owned systems** integrated with the solution. Each external API displays:

- API/System Name
- Integration Type (REST, SOAP, SAML, OIDC, SMTP, etc.)
- Base URL / Endpoint Pattern
- Purpose (e.g., CIF Search, Document Parsing, SSO Authentication)
- Example Endpoints

The catalogue highlights:

- Endpoint Paths
- Supported HTTP Methods
- Functional Purpose
- Input Requirements
- Output Structure
- Authentication/Headers
- Dependencies
- External System Mapping

7. Deployment Mechanism

The system supports modern deployment methodologies that ensure reliable and repeatable releases.

A. Deployment Architecture

- **3-Tier Architecture**
 1. Frontend (Angular)
 2. Backend (Spring Boot)
 3. Database (MS-SQL/Oracle)

B. Deployment Models Supported

- On-Premises Deployment (Preferred by CBK)
- Cloud Deployment (Optional)
 - Private cloud / Containerized
 - Docker + Kubernetes-enabled (optional)

INFRASTRUCTURE REQUIREMENTS

A. Hardware Requirements

(Indicative, can be adjusted based on load)

Layer	Specification	Recommendation
Frontend Server		
Backend Server		
Database Server		
File Storage		

B. Operating Systems

- **Windows Server** – Recommended for ADS integrations

C. Network & Connectivity

- HTTPS enabled
- SSL certificate (internal or public CA)
- Secure VPN/IP Whitelisting for external integrations
- Firewall rules for API communication

8. Software Pre-requisites

Will be shared by the KiyaAI Team during the Deployment Phase.

8.1. ENVIRONMENT SETUP

The solution will run across 3–4 environments:

Environment	Purpose
DEV	Developer testing
TEST / QA	Internal testing + automation
UAT	Client validation (CBK)
PRODUCTION	Live environment

Each environment mirrors production as closely as possible.

9. Project Documentation

The Project Documentation section lists all key documents that support the development, implementation, and ongoing use of the **KiyaAI Credit Assessment System**. These documents ensure that stakeholders—from business users to technical teams—have the necessary clarity for system understanding, maintenance, and onboarding.

A. Functional Specification Document (FSD)

Purpose

The FSD provides a **detailed functional description** of every module, workflow, and business rule defined in the BRD.

It translates business requirements into actionable functional behaviours for development teams.

Contents

- Module-level functional descriptions
- Field-by-field specifications
- User interactions and system responses
- Workflow logic (including maker-checker, approvals, and validations)
- Error messages and exception handling
- Business rules, scoring logic, ratio formulas
- Integration specifications with CIF/LOS/OCR/ADS
- UI wireframes (if applicable)

B. User Manual

Purpose

The User Manual guides end-users in operating the system effectively. It includes navigation flows, screenshots, and instructions for daily activities.

Contents

- Login instructions (Manual + ADS)
- Using the Dashboard
- Creating and managing assessments
- Uploading financial statements (OCR/Excel)
- Ratio interpretation
- Viewing and generating reports
- Template and parameter management

- Admin activities (User, Roles, RBAC, Lookups)
- Troubleshooting and FAQs

Audience

- Analysts
- Reviewers
- Administrators
- Risk & Credit Teams
- Business Users

C. Deployment Guide

Purpose

The Deployment Guide provides step-by-step instructions for setting up and deploying the system in different environments.

Contents

- Prerequisites
- Required software and tools
- Server configuration steps
- Database setup (Oracle schemas, tables, seeds)
- Environment variable configuration
- Email and OCR configuration
- How to deploy frontend & backend builds
- Log configuration and monitoring setup
- Rollback strategies
- Post-deployment validation checklist

Audience

- IT Infrastructure Teams
- DevOps Teams
- System Administrators

—————**END OF THE DOCUMENT**—————